

Teacher Guide — middle / module-12- movement-of-sun

IKS Curriculum

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Overview

Module 12 — Movement of the Sun

Age band: Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min (≈ 2 weeks)

Learning objectives

By the end of this module, students will be able to:

1. **Describe** the Sun’s apparent annual path along the ecliptic (*krānti-vṛtta*) and explain how Earth’s ~23.5° axial tilt produces the seasons.
2. **Distinguish** *uttarāyaṇa* (northward course) from *dakṣiṇāyaṇa* (southward course) and identify the two solstices and two equinoxes (*viṣuva*) on a calendar.
3. **Recite** the 12 *rāśis* (Meṣa through Mīna) in order and locate the Sun approximately in each *rāśi* during the calendar year.
4. **Define** *saṁkrānti* as the Sun’s transit from one *rāśi* to the next, and explain why **Makara Saṁkrānti** (~14 January) and **Karka Saṁkrānti** (~16 July) are tied to the solstices in the traditional reckoning.
5. **Compare** *sāyana* (tropical) and *nirayana* (sidereal) reference frames, and state that the gap between them — the *āyanāṁśa* — is now ~24° because of precession.
6. **Map** the Indian framework to its cross-cultural cousins (Greek zodiac, Egyptian solar calendar, Gregorian calendar) without claiming priority either way.

NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems with mainstream science (here: astronomy and Indian *jyotiṣa* together).
- **§4.23** — Experiential learning: solar-position observation, torch-and-globe demo, paper Sun-path diagrams.
- **§11.8** — Multidisciplinary linkage: geography (latitude, tilt), physics (orbital motion), Sanskrit (terminology), history of science.
- **§4.27** — Local context: students record their own sunrise/sunset times and shadow lengths.

Prerequisites

- Familiarity with the four cardinal directions (N, S, E, W).
- Awareness that the Earth rotates (day/night) and revolves around the Sun (year). NCERT Class 6 Geography Chapter 3 covers this.
- Optional: prior exposure to Module 6 (Moon Phases & Tithi). Useful but not required.

Key vocabulary

See `_shared/glossary.md` for full entries. New terms introduced this module:

- *Sūrya* — the Sun
- *krānti-vṛtta* — the ecliptic
- *ayana* — half-year course (northward / southward)
- *uttarāyaṇa* — northward course
- *dakṣiṇāyana* — southward course
- *viṣuva* — equinox
- *rāśi* — sign of the zodiac (12 of them)
- *saṁkrānti* — the Sun's crossing into a new *rāśi*
- *sāyana* / *nirayana* — tropical / sidereal reference frames
- *ayanāṁśa* — the offset between the two frames (~24° today)

Materials checklist (whole module)

- Globe (any size) + a strong torch or table lamp
- A compass (magnetic or a phone app)
- A 30 cm vertical stick (gnomon) for the schoolyard
- A printable year-calendar with dates marked
- Chart paper / printable circle (for the *rāśi*-wheel)
- Worksheets in the activity folder
- Sunglasses for ANY observer who steps outside — **no direct looking at the Sun**

Day-by-day overview

Day	Title	Focus	Assessment
1	What is the Sun Doing?	Apparent path on the sky	—
2	Why Earth has Seasons	Axial tilt demo	Exit ticket
3	Uttarāyaṇa & Dakṣiṇāyana	The two solstices	Formative quiz Part A
4	The Equinoxes (Viṣuva)	Day and night equal	Exit ticket
5	The 12 Rāśis	Names, order, durations	Formative quiz Part B

Day	Title	Focus	Assessment
6	Samkrānti	Makara, Karka, and the rest	Discussion
7	Sāyana vs Nirayana	Why dates seem to disagree	Compare-contrast worksheet
8	Project Session 1	Build the <i>rāśi</i> -wheel	Project draft
9	Project Session 2	Cross-cultural comparison	Project draft
10	Final Assessment + Showcase	Summative + presentations	Summative + project

Sources used

See `sources.md` for primary and modern references. Key anchors:

- **Sūrya-siddhānta** — Sanskrit astronomical treatise (compiled c. 5th–10th c. CE) describing solar motion, *ayana*, and *rāśi* divisions.
- **Āryabhaṭīya** of Āryabhaṭa (499 CE) — early Indian treatment of precession-related corrections and spherical astronomy.
- **NCERT Geography Class IX, Chapter 3** (“Drainage” / “Climate”) and **NCERT Physics Class XI, Chapter 8** (Gravitation) — modern grade-appropriate references for axial tilt and orbital motion.

Safety

- **Never look at the Sun directly.** Even a quick glance can cause permanent retinal damage. The shadow-stick observation in Day 1 is the **ONLY** safe way to record the Sun’s position in this module. No solar viewing without certified solar filters.
- Outdoor activity (Day 1) requires a shaded standpoint for the observer; only the gnomon’s shadow is studied.

Project brief (one-line teaser)

Each pair builds a paper *rāśi*-wheel — 12 segments labelled in Sanskrit, English, and Greek-derived names, with the approximate date when the Sun is in each — and explains how axial tilt makes the seasons. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 12 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	What is the Sun Doing?	days/day-01-suns-path.md	Shadow-stick observation
2	Why Earth has Seasons	days/day-02-axial-tilt.md	Torch-globe demo + exit ticket
3	Uttarāyaṇa & Dakṣiṇāyana	days/day-03-ayana.md	Calendar marking; formative quiz Part A
4	The Equinoxes (Viṣuva)	days/day-04-equinox.md	Day-length graph
5	The 12 Rāsis	days/day-05-rashis.md	Rāsi-wheel introduction; formative quiz Part B
6	Saṁkrānti	days/day-06-sankranti.md	activities/activity-02-makara-sankranti.md
7	Sāyana vs Nirayana	days/day-07-sayana-nirayana.md	Compare-contrast worksheet
8	Project Session 1: Build	days/day-08-project-session-1.md	Project draft
9	Project Session 2: Refine	days/day-09-project-session-2.md	Project draft
10	Final Assessment + Showcase	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- Day 1's shadow-stick observation needs a sunny mid-morning window. If your timetable lands the lesson in early morning or after a rain, move the outdoor part to a later day and use the time for the explanation.
- The torch-and-globe demo (Day 2) works best with the room blackout-curtained. If your room doesn't black out, demonstrate the day before in a darker space and take photos to project.
- Day 7 (*sāyana* / *nirayana*) is intellectually the densest. Don't squeeze it.
- Project work (Days 8–9) assumes pair vs. individual is decided by Day 5.

Suggested running theme

Open each class with the same prompt on the board: “**Where was the Sun this morning when you left for school?**” (Direction: east-ish — but more north of east in summer, more south of east in winter.) Two students share for 30 seconds each. By Day 10 the class will have a felt sense that the Sun does not rise from the exact same spot every day.

Formative vs. summative weight

- Formative quizzes (Days 3 and 5) are **not graded** — they are diagnostic. Use them to decide who needs scaffolding on Day 7.
- Summative quiz (Day 10) — see rubric in `assessments/rubric.md`. Combined with project, total 40 marks.

What students leave the module with

- A completed paper *rāśi*-wheel with their own labelling.
- The 12 Sanskrit *rāśi* names with English / Greek-derived equivalents.
- A diagram in their notebook showing axial tilt and the four turning points of the year.
- A felt sense that “what day is *Makara Saṁkrānti*?” and “what day is the winter solstice?” are answers to two slightly different questions.

Teacher Notes

Teacher Notes — Module 12 Movement of the Sun (Middle)

Posture for this module

Lead with what students can observe directly: the Sun rises in the east-ish, sets in the west-ish, but the exact spot on the horizon shifts north and south through the year. The Indian terminology (*uttarāyaṇa*, *dakṣiṇāyaṇa*, *rāśi*, *saṁkrānti*) names what you can see, then provides a framework for calculation. Modern astronomy explains the underlying mechanism — Earth’s axial tilt and orbital motion. Both descriptions are about the same sky. The module trains the habit of holding the observational vocabulary and the mechanistic explanation together.

If a student asks “but isn’t astrology nonsense?” — distinguish between *jyotiṣa* as **observational astronomy** (which is what this module is about: tracking solar position, predicting *saṁkrānti* dates, computing the calendar) and *jyotiṣa* as **horoscope prediction** (claims about how planetary positions affect individual lives — outside the science scope of this module). Indian astronomical texts contain both; we teach the first.

Common misconceptions to watch for

1. “**The Sun goes around the Earth.**” Geocentric appearance is real; we observe the Sun’s *apparent* motion. Earth orbits the Sun (heliocentric reality). The two descriptions are tied

by relative motion. Don't shame the "Sun moves" framing — it's the framing the Sūrya-siddhānta and Aristotle and Ptolemy all used. Indian astronomers from Āryabhaṭa onward did consider Earth's rotation; the heliocentric model was finally pinned down in Europe in the 16th–17th century.

2. **"India has summer in winter because it's in the south."** No. The Northern Hemisphere has its summer when the *north* end of Earth's axis tilts toward the Sun, which happens in June. In June it's hot across most of India. India's summer = June, not December.
3. **"Makara Saṁkrānti is the winter solstice."** Close, but not exactly. In the *sāyana* (tropical) frame, the winter solstice is ~21 December. In the *nirayana* (sidereal) frame, the Sun enters *Makara* around 14 January. The two used to coincide ~1700 years ago, but precession has spread them apart by ~24°. Day 7 unpacks this. **Don't fudge it** — the apparent contradiction is the whole interest of the topic.
4. **"Sāyana is right and nirayana is wrong" (or vice versa).** Neither is "right." They are two different reference frames. *Sāyana* (tropical) is fixed to the equinoxes — useful for the seasons. *Nirayana* (sidereal) is fixed to the stars — useful for long-term astronomical observation. Modern astronomy uses *sāyana* for civil calendars and *nirayana* for star catalogues; Indian traditional almanacs (*pañcāṅgas*) use *nirayana* for festival dates.
5. **"The Sun is in a constellation matching my horoscope sign on my birthday."** It is in the *sāyana* sign matching popular Western horoscope columns (those use the tropical zodiac), but the Sun is NOT visually in front of that constellation on that date — because of precession. This is a common eye-opener; let students discover it on Day 7.
6. **"Each rāśi is 30 days long."** Roughly, but not exactly. Because Earth's orbit is elliptical, the Sun's apparent angular speed varies — so some *rāśi* transits last ~29 days, others ~31. The mean is 30.44 days (365.25 / 12).

On Sanskrit pronunciation

Drill these terms aloud across the module — they are not familiar to most students:

- *uttarāyaṇa* — "oo-ttuh-RAH-yuh-nuh" (the ṇ is retroflex; tongue tip touches the back of the palate)
- *dakṣiṇāyana* — "duck-SHI-nah-yuh-nuh" (kṣ is "ksh" or "ksh-ya")
- *saṁkrānti* — "sung-KRAHN-tee" (the ṁ is a nasal hum, mostly anusvāra)
- *viṣuva* — "VI-shoo-vuh"
- *rāśi* — "RAH-shi" (ś is the "sh" of "ship")
- *sāyana* / *nirayana* — "SAH-yuh-nuh" / "ni-ruh-YUH-nuh"
- *ayanāṁśa* — "uh-yuh-NAHM-shuh"
- *krānti-vṛtta* — "KRAHN-ti-VRIT-tah"

Why this module is hard to teach

The *sāyana* / *nirayana* distinction (Day 7) is the genuine intellectual challenge. Many adults — even teachers — quietly carry the wrong idea that *Makara Saṁkrānti* IS the winter solstice. Day

7 is engineered to surface and resolve this. Plan for one student per class to push back hard (“then the calendar must be wrong!”). It isn’t wrong; it answers a different question. The traditional calendar tracks the Sun’s position **against the fixed stars**; the modern civil calendar tracks the Sun’s position **against the equinoxes**. Both are internally consistent; they diverge because the equinoxes precess against the stars.

If the class is stuck, draw this diagram: a circle (the ecliptic), a fixed star marker (the *āśvinī* nakshatra), and a moving arrow marker (the equinox). 2000 years ago they overlapped. The arrow has slid back $\sim 24^\circ$. That’s *ayanāṁśa*.

Differentiation hints

- **Tier up (fast finishers):** Compute their own birthday’s Sun position — both *sāyana* and *nirayana* — using a website like *Stellarium* (free desktop app) or *time-and-date.com*. Compare to their newspaper horoscope sign.
- **Tier down (scaffolding):** Reduce the 12 *rāśi* names to 6 (the four solstice/equinox markers plus two others). Drill these in pairs across Days 5 and 6. The remaining six can come in the project work.

Sensitivity notes

- Astrology is a touchy subject. Some families read horoscopes and *pañcāṅgas* daily; others reject astrology as superstition. The classroom frame is neither. We teach the **astronomy** behind the calendar (what does “the Sun is in Makara” mean as an observable fact about the sky) without taking a position on whether the placement determines personality or fortune.
- *Makara Saṁkrānti* is celebrated as Pongal, Bihu, Lohri, Maghi, and Uttarayan across India. Acknowledge this if students bring it up — it’s a calendar festival rooted in the astronomy we’re studying. Don’t turn the lesson into a festival lesson; do honour the cultural connection.

Safety — Sun viewing

Never allow students to look directly at the Sun. Standard solar-eclipse-grade filters are the only safe direct-view tool. For this module, all observation is **shadow-based**: the gnomon’s shadow on a flat surface tells you where the Sun is without you needing to look up. Treat any “I’ll just peek for a second” as a hard NO.

Project oversight (Days 8–10)

- Confirm groups/pairs by end of Day 5.
- On Day 8, every group should leave class with a sketched *rāśi*-wheel.
- On Day 9, intervene if a group is still planning — they need a draft artefact by end of class.
- For Day 10 presentations, use the 3-minute timer strictly.

Assessment grading sequence

1. Mark Day 3 formative quiz immediately so feedback lands by Day 4.

2. Day 5 formative quiz can be marked during Day 6's discussion.
3. Day 10 summative quiz can be marked over the weekend.
4. Project rubric should be shown to students on Day 8 — let them ask questions about it.

If you only have 6 days, not 10

Compress as follows:

- Day 1 (was Day 1)
- Day 2 (was Day 2)
- Day 3 (was Days 3 + 4 condensed — solstices and equinoxes together)
- Day 4 (was Day 5 — *rāśis*)
- Day 5 (was Days 6 + 7 condensed — *saṁkrānti* and the *sāyana/nirayana* distinction; this is the hard day)
- Day 6 (was Day 10 — summative quiz + show-and-tell, no separate project days)

Skip Day 8–9 project sessions; assign the project as homework instead.

Scientific-religious framing

Uttarāyana and *dakṣiṇāyana* are mentioned in the Mahābhārata (Bhīṣma chooses to die during *uttarāyana*, considered auspicious) and elsewhere in the Sanskrit corpus. We acknowledge this in passing — it shows that the astronomy was part of the lived culture — but we **do not teach the religious significance** in class. The astronomy is what's testable; the religious framing belongs to a different subject. Be ready to redirect: “we'll learn what *uttarāyana* means astronomically here; if you want to discuss why it was considered auspicious, that's a Sanskrit-literature conversation, not a science one.”

Sources

Sources — Module 12 Movement of the Sun (Middle)

This module does not use RAG-retrieved citations (the vidya-karana corpus does not cover *jyotiṣa*). All references below are real published works; only paraphrased material is taken from them, no invented quotations.

Primary Sanskrit sources (paraphrased only, not quoted)

Source	Used in	What it contributes
Sūrya-siddhānta (Sanskrit astronomical compendium; modern critical editions ~5th–10th c. CE in	Days 3, 5, 6, 7	The classical Indian treatment of solar motion, the <i>rāśi</i> divisions, <i>uttarāyana</i> / <i>dakṣiṇāyana</i> , and the framework for calculating <i>saṁkrānti</i> dates.

Source	Used in	What it contributes
their current form)		
Āryabhaṭīya of Āryabhaṭa (499 CE)	Day 8 (Senior reference)	One of the earliest extant Indian astronomical treatises; treats the Earth as rotating, gives values for the length of the year, and contains the <i>gola</i> (sphere) chapter.
Bṛhat Saṃhitā of Varāhamihira (6th c. CE)	Day 6	Encyclopedic compendium with chapters on the <i>saṃkrāntis</i> and their interpretation.

These works are widely available in scholarly translation (Burgess 1860 for Sūrya-siddhānta; Shukla 1976 for Āryabhaṭīya; M. Ramakrishna Bhat 1981 for Bṛhat Saṃhitā). The classroom does not need student-facing direct quotes for Middle band — a paraphrase with the title and a translator’s reference is sufficient.

Citation rule reminder (Middle band): at least ONE paraphrase with reference appears in days/day-05-rashis.md and days/day-06-sankranti.md. We do not quote any specific verse number we cannot personally verify; paraphrasing the text’s overall content is fine.

Modern science references

- **NCERT Class 6 Geography, Chapter 3** — “Motions of the Earth.” Introductory treatment of rotation, revolution, axial tilt, and the seasons.
- **NCERT Class 9 Geography, Chapter 4** — “Climate.” Touches the Tropic of Cancer / Tropic of Capricorn and seasonality across India.
- **NCERT Class 11 Physics, Chapter 8** — “Gravitation.” Background on planetary orbital motion.
- **NASA’s solar-system educational pages** (“Solstices and Equinoxes,” “How Earth’s Tilt Causes Seasons”) — useful classroom diagrams. URLs change; search by topic name on the NASA Science Education domain.

Historical context references (teacher reading, optional)

- Plofker, K. (2009). *Mathematics in India*. Princeton University Press. — Sets the historical context of Indian astronomy alongside other traditions; chapter on Āryabhaṭa, Brahmagupta, and the Sūrya-siddhānta.
- Pingree, D. (1981). *Jyotiḥśāstra: Astral and Mathematical Literature*. A History of Indian Literature vol. 6. Otto Harrassowitz. — The reference catalogue of Indian astronomical and astrological texts; describes the *rāśi* system in its historical context.
- Subbarayappa, B. V. & Sarma, K. V. (1985). *Indian Astronomy: A Source-Book*. Nehru Centre, Bombay. — Translations of key passages from Indian astronomical texts.

These are reference works for the teacher, not assigned to students.

On the “sidereal vs tropical” question (Day 7)

The current *ayanāṁśa* (gap between *sāyana* and *nirayana* zodiacs) is approximately 24° (Lahiri *ayanāṁśa*, the value adopted by the Government of India in its civil almanac). This corresponds to the fact that ~1700 years have passed since the two coincided. The drift rate is ~50.3 seconds of arc per year (about 1° per 72 years) due to the precession of the Earth’s rotation axis — a 25,800-year cycle.

These are standard astronomical values, available in any modern positional-astronomy textbook (e.g. Smart, W. M., *Textbook on Spherical Astronomy*, Cambridge UP).

What we will NOT cite

- We do not cite specific verse numbers from the *Sūrya-siddhānta* or *Āryabhaṭīya* in student materials. The numbering varies across editions and translations; without RAG verification we paraphrase only.
- We do not cite any Purāṇic verse for *uttarāyaṇa* / *dakṣiṇāyaṇa* in student materials. These terms appear in many texts; nothing student-facing needs a verse-by-verse anchor at Middle band.

Day 1 — What is the Sun Doing?

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will describe the Sun’s daily and annual apparent motion and articulate the question: *does the Sun rise from exactly the same spot every day?*

Materials

- Whiteboard
- A 30 cm wooden dowel or pencil + a piece of cardboard / paper to plant it on (the gnomon)
- A compass (or a phone with a compass app)
- A printout of a year-calendar (Jan to Dec, all dates visible) — one per pair
- Optional: a photo of the room taken at the same time on different days (a teacher prep item)

Vocabulary introduced today

- *Sūrya* (IAST: *sūrya*; lit. “the Sun”) — the luminary at the centre of our solar system; the object we will track for the next 10 days.
- *krānti-vṛtta* (IAST: *krānti-vṛtta*; lit. “deflection-circle”) — the **ecliptic**, the apparent path the Sun traces across the sky over a year.

Lesson flow

Warm-up (5 min)

- Prompt on board: “Where was the Sun this morning when you left home?”
- Take 3–4 responses. Most will say “east” or “in front of me / behind me.” Some will say “I didn’t notice.” That’s the honest answer — and the launching point.
- Teacher: “Today we’re going to start noticing. For two weeks we’ll track the Sun. The Indian astronomical tradition has been doing exactly this for 2000+ years.”

Core (20 min)

1. **Two motions, one Sun.** Sketch on the board:

- **Daily motion:** rises east-ish, climbs, sets west-ish. Takes ~24 hours.
- **Annual motion:** the rising spot shifts north (summer) and south (winter) along the horizon, completing a full cycle in ~365.25 days.

2. **The Sun’s apparent path on the sky.** Show this:

NORTH
|
W---+---E ← in March/September the Sun rises due east
| and sets due west
SOUTH

W---+---E ← in June the rising point shifts NORTH of east
(high arc in the sky, long days)

W---+---E ← in December it shifts SOUTH of east
(low arc, short days)

- The path the Sun traces against the *stars* over a year is called the **ecliptic** (*krānti-vṛtta*).
- From Earth, we don’t see the stars in daytime — but Indian astronomers worked out the ecliptic by noting which stars came up just before sunrise and just after sunset.

3. **Two questions for the module.** Write on the board:

- **Q1:** Why does the rising spot shift? (*Answer = axial tilt — Day 2.*)
- **Q2:** What did Indian astronomers call the Sun’s slow north-south swing? (*Answer = uttarāyaṇa and dakṣiṇāyana — Day 3.*)

Activity (15 min) — Shadow-stick observation

If the weather permits and the schoolyard has a sunny spot:

- Plant the gnomon (the dowel) upright in a piece of cardboard. Mark north on the cardboard using the compass.

- At a known time (say, the start of the lesson), trace the shadow of the gnomon onto the cardboard. Write the time and the date next to the line.
- Return to the same gnomon in 10 minutes. The shadow has moved. Trace the new line.
- Discuss: *“The Sun seemed to stand still while you were in class — but the shadow moved. What does that tell you about how the Sun moved?”*

If weather doesn’t permit: do the activity indoors with a strong torch (acting as the Sun) and a pencil-on-paper gnomon. Move the torch slowly along an arc to mimic the daily motion. Each “minute” of torch travel = ~4 minutes of real Sun travel.

Safety reminder: the students look at the **shadow**, not the Sun. Anyone caught glancing up at the Sun pays a forfeit (a fun one — like doing the daily prompt next class).

Wrap-up (5 min)

- Exit prompt: *“In one sentence — how can you tell tomorrow morning whether the Sun has ‘moved north’ or ‘moved south’ since today?”*
- Take 2 responses. Acceptable answer: *“Compare the sunrise spot to a known landmark on the horizon — has the rising point moved closer to north or to south?”*
- Preview Day 2: *“Tomorrow — WHY does the Sun’s spot shift? Bring a globe if you have a small one at home.”*

Student-facing content

The Sun has two motions you can see:

- **Daily** — rises east-ish, sets west-ish. Takes 24 hours.
- **Annual** — the rising spot drifts north and south along the horizon over a year.

The Sun’s yearly path across the stars is called the **ecliptic** (*krānti-vṛtta*). Indian astronomers, like astronomers everywhere, traced this path by watching which stars were just visible at dawn and dusk.

Today, you have noticed the daily motion (via the gnomon). Over the next two weeks, you’ll learn the names for the annual motion: *uttarāyaṇa* (northward course) and *dakṣiṇāyana* (southward course).

Homework

- For three mornings this week, when you leave home, glance at the eastern horizon. Note where exactly the Sun rises relative to a landmark (a tree, a building, a flagpole). Sketch the spot on a small map — three sketches.
- Bring the calendar printout to Day 2.

Differentiation

- **One tier up:** Find out (online or in the school library) the **time** of sunrise and sunset for today AND for one day exactly six months from now. By how many minutes does the day length differ?
- **One tier down:** Just do the morning-glance — don’t worry about sketching maps. Three mental notes are enough.

Teacher notes

- Some students will know the Sun-rises-in-the-east rule and stop there. Push: “*Always east? Or east-ish?*” The “ish” is the entire module.
- Don’t introduce axial tilt yet. That’s Day 2. Let students sit with the *observation* of the shifting spot before getting the *explanation*.
- The shadow activity is more valuable than the explanation. If you only have 30 minutes, run the activity and trim the Core.

Day 2 — Why Earth has Seasons

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will explain that Earth’s $\sim 23.5^\circ$ axial tilt — not its distance from the Sun — is what causes the seasons, and demonstrate this with a torch-and-globe model.

Materials

- A small globe (or a ball with N and S marked) — one per group ideally, one as demo minimum
- A strong torch / table lamp (the “Sun”)
- A protractor (to verify the tilt angle)
- A blackout / curtained room if possible, OR a hallway
- Student notebooks

Vocabulary introduced today

- **Axial tilt** (obliquity of the ecliptic) — Earth’s rotation axis is tilted $\sim 23.5^\circ$ from a line perpendicular to its orbital plane. This tilt does not change over a year — Earth keeps its tilt pointed in the same direction in space as it goes around the Sun.

(No new Sanskrit term today — yesterday’s *krānti-vṛtta* covers it. The axial tilt is what *causes* the ecliptic to be a *circle* on the sky rather than just running along the equator.)

Lesson flow

Warm-up (5 min)

- Prompt: “Why is it hot in May–June in India and cool in December–January?”
- Likely wrong answer (very common): “Because Earth is closer to the Sun in summer and farther in winter.”
- Surprise: Earth is actually slightly *closer* to the Sun in **January** (perihelion ~ 3 January, 147.1 million km) and slightly *farther* in **July** (aphelion ~ 4 July, 152.1 million km). India is hot in May–June anyway. So distance is not the answer.
- Real question: “What else could cause seasons?”

Core (20 min)

1. The torch-and-globe demo. Set up:

- Place the torch on a desk pointing horizontally.
- Hold the globe at arm's length, with its axis tilted $\sim 23.5^\circ$ (use the protractor to check). The N pole should lean either toward or away from the torch.

Move the globe through four positions, each $\sim 90^\circ$ around the torch, keeping the axis tilt **fixed in space** (the N pole always points the same direction):

Position	What it represents	Northern Hemisphere	Southern Hemisphere
N pole tilted TOWARD torch	June solstice	Summer (long days)	Winter (short days)
N pole tilted SIDEWAY S (1)	September equinox	Day = night	Day = night
N pole tilted AWAY from torch	December solstice	Winter (short days)	Summer (long days)
N pole tilted SIDEWAY S (2)	March equinox	Day = night	Day = night

2. What students should see:

- When N is tilted toward the torch: the upper half of the globe is brightly lit, and the lit region extends past the N pole. India is well-lit and the Sun is high in the sky. → Summer.
- When N is tilted away: the upper half is poorly lit, the lit region falls short of the N pole. India is lit at a shallow angle. → Winter.

3. Two things the tilt does:

- **Day length** — when tilted toward, the daytime arc through India is *longer* than the night-time arc. → Longer days in summer.
- **Angle of sunlight** — sunlight hits a tilted surface at a steeper angle in summer (more concentrated, more heat per square metre) and a shallow angle in winter (more spread out, less heat).

4. Why does the tilt stay fixed? Earth's rotation acts like a spinning top — it resists changes in its tilt direction. Over very long times ($\sim 25,800$ years) the tilt direction wobbles in a slow circle (this is **precession** — Day 8 senior band; we hint at it here only as a footnote).

Activity (15 min) — Build your own tilt model

In pairs, students build a paper “season-checker”:

- Take an A4 sheet. Mark a small circle in the centre — that’s the Sun.
- Draw Earth’s orbit as a big circle around it.
- At four positions on the orbit (top, right, bottom, left), draw a small Earth with its axis tilted to the **right** of vertical (same direction in all four).
- Mark which position is summer in the Northern Hemisphere, which is winter, which are the two equinoxes.
- Label each: June / September / December / March.
- Bonus: mark the Indian subcontinent on each Earth and shade where it’s day.

Wrap-up (5 min)

- **Exit ticket on a slip of paper:** “Which is hotter in India in June: the *day length* or the *angle of sunlight* on your skin? Or both? In one sentence.”
- Acceptable answer: BOTH — longer days mean more total hours of sunlight, AND the steeper angle means each minute of sunlight delivers more heat per square metre.

Homework

- On the calendar printout, mark the four dates: **~21 March (March equinox), ~21 June (June solstice), ~23 September (September equinox), ~21 December (December solstice).**
- Bring the marked calendar to Day 3.

Student-facing content

Earth is tilted. Its rotation axis is at $\sim 23.5^\circ$ from straight-up (relative to its orbit around the Sun). And the tilt direction *doesn’t change* as Earth orbits.

This single fact causes the seasons:

- When the N pole leans TOWARD the Sun ($\sim$ June 21) → summer in India.
- When it leans AWAY ($\sim$ December 21) → winter in India.
- When it leans SIDEWAYS ($\sim$ March 21 and $\sim$ September 23) → day = night everywhere on Earth.

Surprise fact: Earth is actually *closer* to the Sun in January than in July. Distance is not what causes seasons. Tilt is.

Differentiation

- **One tier up:** What would the seasons be like if Earth’s tilt were 0° ? (No seasons; equal day/night everywhere all year; equatorial regions hot, polar regions cold permanently.) What if it were 45° ? (Extreme seasons — the Arctic Circle would extend down to Delhi’s latitude in summer.)
- **One tier down:** Focus on the demo. Just remember: “tilted toward = summer, tilted away = winter, sideways = equinox.”

Teacher notes

- The “Earth closer to Sun = summer” misconception is *very* persistent. Highlight that India’s June is summer despite Earth being farthest from the Sun in July. The misconception breaks instantly when this is named.
- Some students will ask: “*Then why is India summer in June and not in winter like Australia?*” — Yes, India is in the Northern Hemisphere; Australia is in the Southern. Same Earth tilt, opposite seasons. (Australia is wearing shorts in December, jumpers in June.)
- The demo will fail visually if the room is too bright. If you can’t darken the room, use a smaller globe and bring the torch closer (3–4 cm from the globe surface).

Citation(s) used in this lesson

- Modern axial-tilt and orbital values from standard NCERT-level astronomy. No primary-source citation needed at this band.

Day 3 — Uttarāyana and Dakṣiṇāyana

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will name the two solstices, identify the half-years they bookend, and use the Sanskrit terms *uttarāyana* and *dakṣiṇāyana* correctly.

Materials

- The calendar from Day 2 (with the four turning-points marked)
- The whiteboard
- A printable two-circle handout (one circle per half-year — see Activity)

Vocabulary introduced today

- *ayana* (IAST: *ayana*; lit. “course, path”) — a half-year course of the Sun, either northward or southward.
- *uttarāyana* (IAST: *uttara-āyana*; lit. “northern course”) — the half-year from the winter solstice (~21 December) to the summer solstice (~21 June). The Sun’s noon altitude *rises* in the Northern Hemisphere over this period.
- *dakṣiṇāyana* (IAST: *dakṣiṇa-āyana*; lit. “southern course”) — the other half-year, ~21 June to ~21 December. The Sun’s noon altitude *falls* in the Northern Hemisphere.
- **Solstice** — Latin *sol-stat*, “Sun stands still.” On the solstice day, the Sun’s noon altitude reaches its highest or lowest value of the year, then begins reversing direction. Sanskrit name: *ayana-samkrānti*.

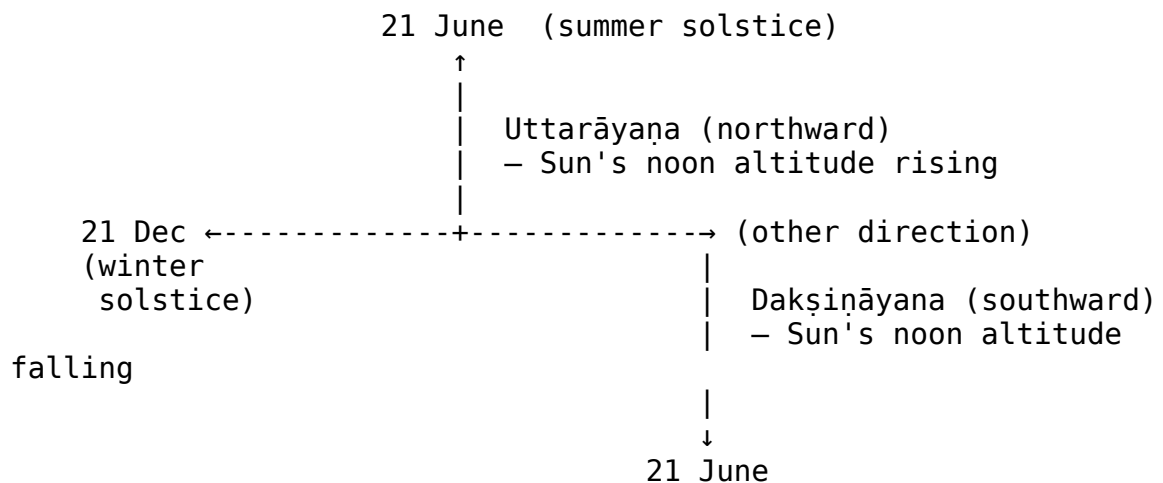
Lesson flow

Warm-up (5 min)

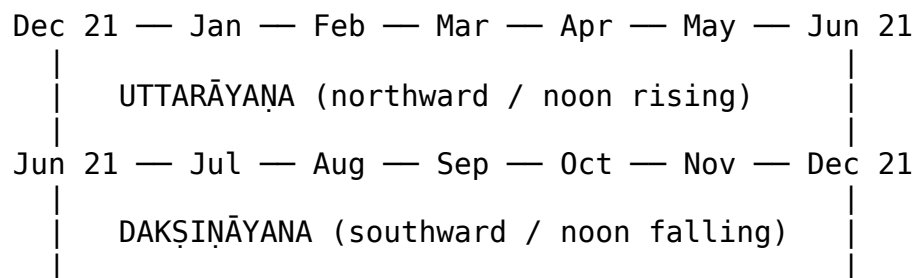
- Open the calendar. The four marks from yesterday — March 21, June 21, September 23, December 21 — are visible.
- Prompt: “Today (insert today’s date) — which ‘turning point’ is closest? And which side of it are we on?”
- Take 2 responses. They’ll likely give the right answer with help.

Core (20 min)

1. **Two halves of the year.** Sketch on the board:



No, draw it as a clean cycle:



2. **What does “northward” mean exactly?**

- The Sun’s **noon position** in the southern sky shifts **NORTH** a little each day during *uttarāyana*.
- In northern India (e.g. Delhi, latitude ~28°N), the noon Sun is **very low in the south** in late December and **almost overhead** in late June.
- “Almost overhead” means the noon shadow of a vertical stick is at its shortest.

3. **Paraphrase from the Sūrya-siddhānta** (Middle-band citation requirement):

The Indian astronomical text **Sūrya-siddhānta** (compiled in its current form between the 5th and 10th centuries CE) describes the year as having two *ayanas* — *uttarāyana* and *dakṣiṇāyana* — and gives mathematical formulas for the Sun’s daily motion in each. — paraphrased from Sūrya-siddhānta, chapter on solar motion (see [sources.md](#))

4. Tropic of Cancer and Tropic of Capricorn (linking to Geography):

- The latitude at which the Sun is *directly overhead at noon on the June solstice* = 23.5° N = **Tropic of Cancer**. (Cancer = the Latin/Greek name for *Karka rāśi*. The Sun used to be in Karka at the June solstice — Day 7 will explain why “used to be.”)
- The latitude at which the Sun is directly overhead on the December solstice = 23.5° S = **Tropic of Capricorn**. (Capricorn = *Makara*. Same caveat.)
- India straddles the Tropic of Cancer — it runs roughly through Bhopal, Ranchi, and Kolkata.

Activity (15 min) — Calendar marking and formative quiz Part A

Part 1 (8 min) — calendar marking:

- On their marked calendar, students label each half:
 - From Dec 21 → Jun 21: write “**UTTARĀYANA**” in colour
 - From Jun 21 → Dec 21: write “**DAKṢIṆĀYANA**” in colour
- They also write the Sanskrit names next to the four turning-points:
 - Dec 21: *Uttarāyana begins (winter solstice)*
 - Jun 21: *Dakṣiṇāyana begins (summer solstice)*
 - Mar 21 / Sep 23: *Viṣuva* — we’ll meet these tomorrow.

Part 2 (7 min) — Formative Quiz Part A: see quizzes/formative.md. Collect.

Wrap-up (5 min)

- Exit prompt: “*Today (today’s date) — are we in uttarāyana or dakṣiṇāyana? Justify in one sentence.*”
- Preview Day 4: “*Tomorrow — the equinoxes. The days when day and night are equal.*”

Student-facing content

The Sun’s year has two halves:

Half	Sanskrit name	When	What the Sun does (Northern Hemisphere)
North ward	<i>uttarāyana</i>	~21 Dec → ~21 Jun	Noon altitude rises; days lengthen
South ward	<i>dakṣiṇāyana</i>	~21 Jun	Noon altitude falls; days shorten

Half	Sanskrit name	When	What the Sun does (Northern Hemisphere)
		→ ~21 Dec	

The turning days are the **solstices** (in Sanskrit, *ayana-saṁkrāntis*).

The latitudes where the Sun is directly overhead at the two solstices have names — **Tropic of Cancer** (23.5° N, on the June solstice) and **Tropic of Capricorn** (23.5° S, on the December solstice). India straddles the Tropic of Cancer.

Homework

- Pick a vertical object near your home that you can see at noon (a flagpole, a tree, a building). For two days this week, at exactly 12:00 noon, look at its shadow. Is the shadow getting *shorter* or *longer*? In a sentence: which *ayana* are we in, and does your shadow observation match?

Differentiation

- **One tier up:** Indian latitude: ~8° N (Kanyākumāri) to ~35° N (Srinagar). For a place at 30° N, how high (in degrees from the horizon) is the noon Sun on the June solstice? On the December solstice? Hint: $90^\circ - \text{latitude} \pm 23.5^\circ$.
- **One tier down:** Focus on the two names. *Uttara* = north. *Dakṣiṇa* = south. *Ayana* = course. That's enough for today.

Teacher notes

- Don't conflate *uttarāyana* with the festival of *Makara Saṁkrānti* yet. Today's *uttarāyana* starts on **December 21** (the modern *sāyana* solstice). *Makara Saṁkrānti* falls on ~**January 14** (the *nirayana* sidereal reckoning). They are off by ~24 days. Why? Day 7 unpacks it. Just hold the puzzle for now if a student raises it.
- The *Sūrya-siddhānta* paraphrase satisfies the Middle-band citation requirement for the module.

Citation(s) used in this lesson

- **Sūrya-siddhānta** (Sanskrit astronomical compendium, current critical text 5th–10th c. CE) — paraphrase of the two-*ayana* division of the solar year. See *sources.md* for the full reference.

Day 4 — The Equinoxes (Viṣuva)

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will define an equinox as the day when day and night are equal, identify the two equinoxes in the year, and explain — in terms of axial tilt — why the days are equal on those two dates.

Materials

- The Day 2 torch-globe setup
- A printable day-length graph (x-axis = months, y-axis = day length in hours; blank for the students to fill)
- Student notebooks

Vocabulary introduced today

- *viṣuva* (IAST: *viṣuva*; lit. “equal, balanced”) — an **equinox**, one of the two days a year when day and night have equal length everywhere on Earth (~12 hours each). Two of them: the **vernal/March equinox** (~21 March) and the **autumnal/September equinox** (~23 September).
- *viṣuvad-dina* — another classical name, lit. “the equal day.”

Lesson flow

Warm-up (5 min)

- Prompt: “What does ‘equinox’ literally mean?” — break it down: equi (equal) + nox (night). Latin for “equal night.” The Sanskrit *viṣuva* means the same.
- Recall yesterday’s diagram. Today’s focus is on the two SIDEWAYS positions of Earth (March and September).

Core (20 min)

1. **The geometry of an equinox.** Set up the torch-globe again:
 - Tilt the globe so the N pole leans neither toward nor away from the torch — it’s “sideways.”
 - Notice the **terminator** (the line between light and dark on the globe) — it now passes through both poles, dividing Earth into two equal hemispheres of day and night.
 - **Consequence:** every place on Earth gets 12 hours of day and 12 hours of night.
2. **Why “equal” everywhere?** The day length at any latitude depends on what fraction of that latitude-line is in daylight. On the equinox, the day/night terminator passes through both poles — so every latitude-line is exactly half lit, half dark.
3. **Two equinoxes per year:**

Date	Name	What happens after
~21 March	Vernal / spring equinox (<i>vasanta-viṣuva</i>)	Days get longer in N hemisphere → spring → summer
~23 September	Autumnal / autumn equinox	Days get shorter in N hemisphere → autumn → winter

Date	Name	What happens after
	(śarad-viṣuva)	

4. **Two equinoxes + two solstices = four turning points of the year.** Draw on board:

~21 June (summer solstice)
 * Sun highest at noon, longest day
 (turning point – uttarāyaṇa ends, dakṣiṇāyaṇa begins)
 /
 ~23 Sept (autumn equinox / śarad-viṣuva)
 day = night everywhere
 /
 ~21 Dec (winter solstice)
 * Sun lowest at noon, shortest day
 (turning point – dakṣiṇāyaṇa ends, uttarāyaṇa begins)
 /
 ~21 Mar (spring equinox / vasanta-viṣuva)
 day = night everywhere

5. **A note on the dates:** The dates shift by ± 1 day depending on which year (leap years, the slight unevenness of Earth's orbit). Modern almanacs give exact dates and times.

Activity (15 min) — The day-length graph

In pairs, students fill a blank graph:

- x-axis: months January to December
- y-axis: day length in hours (8 to 14)

The teacher provides a small table of approximate day lengths in Delhi (latitude 28.6° N):

Month	Day length (Delhi)
Jan	~10 h 25 min
Feb	~11 h 05 min
Mar	~12 h 00 min
Apr	~12 h 50 min
May	~13 h 30 min
Jun	~13 h 50 min
Jul	~13 h 40 min
Aug	~13 h 00 min
Sep	~12 h 10 min
Oct	~11 h 25 min
Nov	~10 h 40 min
Dec	~10 h 15 min

Students plot the points, connect them, and mark:

- The **two highest points** (close together in June) — summer solstice.
- The **two lowest points** (close together in December) — winter solstice.
- The **two crossings of the 12-hour line** — the equinoxes.

Discussion (last 3 min):

- Why is the difference between longest and shortest day only ~3.5 hours in Delhi, but ~16 hours in London (latitude 51° N)? (*Answer: the higher the latitude, the more extreme the day-length swing.*)
- What would this graph look like at the equator? (*Answer: nearly flat at 12 h all year.*)
- At the North Pole? (*Answer: 24 h of daylight in summer, 24 h of darkness in winter.*)

Wrap-up (5 min)

- Exit ticket: “Why are equinoxes called ‘equal’? Answer in one sentence.”
- Preview Day 5: “Tomorrow — the 12 *rāsis*. The Sun moves through one each month.”

Student-facing content

Equinox = day equals night.

Sanskrit	Date	Position of Earth	Day length
<i>vasanta-viṣuva</i>	~21 March	tilt sideways	12 h day, 12 h night
<i>śarad-viṣuva</i>	~23 September	tilt sideways the other way	12 h day, 12 h night

Together with the two **solstices** (June and December), these are the four turning-points of the year.

The Sun crosses the celestial equator on each equinox — moving from the Southern celestial hemisphere into the Northern (in March), or vice versa (in September).

Homework

- Compare the sunrise / sunset times in your local newspaper or weather app TODAY with the times exactly six months from now (you can look up future-six-months online or in a printed almanac). Calculate the difference in day length. One paragraph.

Differentiation

- **One tier up:** What is the “true” length of the solar year? Why is it 365.25 days and not 365 exactly? How does the calendar handle this? (Answer involves leap years; the Gregorian system; the Indian *adhikamāsa* in lunisolar context.)
- **One tier down:** Just remember: two equinoxes, two solstices, four turning-points. Mark all four on a calendar.

Teacher notes

- Some students will object: “the newspaper says sunrise was at 6:23, sunset at 18:31 — that’s 12 hours and 8 minutes, not 12 hours exactly.” That’s because the Sun is a **disc** (not a point) and “sunrise” is when the **top** of the disc clears the horizon, and “sunset” is when the **last** of the disc disappears. So there’s a small offset (~5–10 min depending on latitude). Also, atmospheric refraction bends the Sun’s light, making it visible slightly before geometric sunrise. Accept the observation and explain it briefly.
- The day-length graph activity often produces the “aha” moment — students see the year’s shape.

Citation(s) used in this lesson

- Day-length figures are approximate values for Delhi (28.6° N), computable from standard spherical-astronomy formulas. Verifiable in any almanac.

Day 5 — The 12 Rāśis

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will recite the 12 *rāśi* names in order in Sanskrit, match each to its Greek/English equivalent, and identify the approximate calendar month during which the Sun is in each *rāśi* (using the *nirayana* / sidereal system).

Materials

- A large printed circle divided into 12 equal segments (the *rāśi*-wheel template)
- Coloured pencils / sketch pens
- A printed reference table of the 12 *rāśi* names with English translations (from the glossary)
- Student notebooks

Vocabulary introduced today

- *rāśi* (IAST: *rāśi*; lit. “heap, mass; a sign of the zodiac”) — one of 12 thirty-degree-wide segments of the **ecliptic** (the Sun’s annual path on the sky). The Sun is in each *rāśi* for about 30 days.

Lesson flow

Warm-up (5 min)

- Prompt: “What’s your ‘star sign’ according to newspaper horoscopes?” — students call out.
- Teacher: “Those English names — Aries, Taurus, Gemini — they have Sanskrit equivalents. The Indian tradition has 12 *rāśis*, and the Sun visits one each month. Today we learn them.”

Core (20 min)

1. **The ecliptic is divided into 12 equal segments.** Each segment is $360^\circ / 12 = 30^\circ$ of the sky. The Sun crosses each in about 30 days (well, 30.44 on average — see Day 6 footnote).
2. **The 12 *rāśi* names in order.** Write on the board:

#	Sanskrit (IAST)	Greek/Latin/ English	What the symbol depicts
1	<i>Meṣa</i>	Aries	Ram
2	<i>Ṛṣabha</i>	Taurus	Bull
3	<i>Mithuna</i>	Gemini	Twins
4	<i>Karka / Karkaṭa</i>	Cancer	Crab
5	<i>Siṃha</i>	Leo	Lion
6	<i>Kanyā</i>	Virgo	Maiden
7	<i>Tulā</i>	Libra	Scales
8	<i>Ṛścika</i>	Scorpio	Scorpion
9	<i>Dhanus</i>	Sagittarius	Bow / Archer
10	<i>Makara</i>	Capricorn	Sea-goat
11	<i>Kumbha</i>	Aquarius	Water-pot
12	<i>Mīna</i>	Pisces	Fishes

Notice: the Sanskrit and Greek images are *nearly identical*. The bull, twins, crab, lion, scales, scorpion, archer, fishes — same animals. This is not a coincidence; the two systems were in contact during the Hellenistic period (last few centuries BCE) and there was real exchange. Day 9 will compare them in detail.

3. **When is the Sun in each *rāśi*?** Roughly (in the *nirayana* / sidereal system used by traditional Indian almanacs):

<i>Rāśi</i>	Sun is here approximately
<i>Meṣa</i>	14 Apr → 14 May
<i>Ṛṣabha</i>	14 May → 14 Jun
<i>Mithuna</i>	14 Jun → 16 Jul
<i>Karka</i>	16 Jul → 16 Aug
<i>Siṃha</i>	16 Aug → 16 Sep
<i>Kanyā</i>	16 Sep → 17 Oct
<i>Tulā</i>	17 Oct → 16 Nov
<i>Ṛścika</i>	16 Nov → 15 Dec
<i>Dhanus</i>	15 Dec → 14 Jan
<i>Makara</i>	14 Jan → 13 Feb
<i>Kumbha</i>	13 Feb → 14 Mar

Rāṣī

Sun is here approximately

Mīna

14 Mar → 14 Apr

Each crossing is a *saṁkrānti* (Day 6).

4. **Paraphrase from the Sūrya-siddhānta** (re-anchoring the citation):

The Indian astronomical text **Sūrya-siddhānta** divides the ecliptic into 12 equal *rāṣīs* of 30° each and gives the Sun’s daily angular speed — about 0°59’8” per day on average — which corresponds to one full circuit in ~365.25 days. — paraphrased from Sūrya-siddhānta, chapter on the Sun’s motion (see [sources.md](#))

Activity (15 min) — Begin the *rāṣī*-wheel + formative quiz Part B

Part 1 (10 min) — start the wheel:

- Distribute the *rāṣī*-wheel template (a circle divided into 12 wedges, each labelled with a number).
- Students write the Sanskrit name in each wedge in clockwise order (starting from *Meṣa* at the top).
- Below the Sanskrit name, the English / Greek name.
- Below that, the calendar month-range (e.g. *Meṣa* — 14 Apr → 14 May).
- They will finish coloring and decorating during Days 8–9 project work.

Part 2 (5 min) — Formative Quiz Part B: see [quizzes/formative.md](#). Collect.

Wrap-up (5 min)

- Choral recitation: students read the 12 names aloud together, twice.
- Preview Day 6: “*Tomorrow — saṁkrānti. What the Sun does on the day it crosses from one rāṣī to the next. There are ~12 saṁkrāntis per year. The most famous one is **Makara Saṁkrānti** — around 14 January.*”

Student-facing content

The ecliptic is divided into 12 equal segments — the *rāṣīs*.

Each *rāṣī* is 30° of sky. The Sun spends ~30 days in each, and ~365 days to complete the cycle.

The 12 names, in order, are:

Meṣa, Vṛṣabha, Mithuna, Karka, Simha, Kanyā, Tulā, Vṛścika, Dhanus, Makara, Kumbha, Mīna.

They map directly onto the Greek/English: Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius, Pisces.

Homework

- Memorise the 12 *rāṣi* names in order. Use a phone-recording trick: record yourself saying them aloud, listen back twice a day. Test yourself by saying them without looking, in the order Meṣa → Mīna.
- Bonus: which *rāṣi* is the Sun in right now? (Check the *nirayana* table; today's date determines the answer.)

Differentiation

- **One tier up:** The *nirayana* table here is based on the Lahiri *ayanāmśa* (~24° offset from the equinox). If you use the *sāyana* (tropical) system instead — used by Western horoscope columns — the Sun is in *Meṣa* / Aries from **21 March** to **20 April**, not 14 April to 14 May. Why the difference? Day 7 explains. (Hint: precession.)
- **One tier down:** Memorise just 4 names: *Meṣa* (Apr), *Karka* (Jul), *Tulā* (Oct), *Makara* (Jan). These are the four “anchors” — they fall near the solstices and equinoxes. The rest can come over the week.

Teacher notes

- Some students will get confused that their newspaper “Sun sign” doesn't match the *nirayana* dates here. Reassure: this is expected — Western horoscope columns use the *sāyana* system; this table uses *nirayana*. They differ by ~24 days. Day 7 explains.
- The Sanskrit-Greek correspondence is genuinely striking and a good talking point. The historical reason: contact through the Hellenistic world after Alexander's campaigns, with bidirectional flow.

Citation(s) used in this lesson

- **Sūrya-siddhānta** — paraphrase of the 12-*rāṣi* division and the Sun's mean angular speed. See [sources.md](#).

Day 6 — Saṁkrānti

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will define *saṁkrānti* as the Sun's transit from one *rāṣi* to the next, name **Makara Saṁkrānti** and **Karka Saṁkrānti** as the two solstice-related *saṁkrāntis*, and locate all 12 *saṁkrāntis* on a calendar.

Materials

- Yesterday's *rāṣi*-wheel (work-in-progress)
- The marked calendar from Days 2–3
- A list of regional *Makara Saṁkrānti* names (Pongal, Lohri, Bihu, Maghi, Uttarayan)
- Activity 02 worksheet

Vocabulary introduced today

- *saṁkrānti* (IAST: *saṁ-krānti*; lit. “transit, crossing”) — the moment the Sun crosses from one *rāśi* into the next. There are ~12 *saṁkrāntis* per year, one for each *rāśi*-boundary.
- *Makara Saṁkrānti* (IAST: *makara saṁkrānti*) — the *saṁkrānti* around **14 January** when the Sun enters *Makara* (Capricorn). Traditionally marks the start of *uttarāyaṇa* in the *nirayana* reckoning.
- *Karka Saṁkrānti* (IAST: *karka saṁkrānti*) — the *saṁkrānti* around **16 July** when the Sun enters *Karka* (Cancer). Traditionally marks the start of *dakṣiṇāyaṇa* in the *nirayana* reckoning.

Lesson flow

Warm-up (5 min)

- Prompt: “Have you ever celebrated **Pongal, Lohri, Bihu, or Uttarayan?**” Likely many hands. “All of these fall around the same date — January 14. Why?”
- Take 2 responses. Then: “They’re all names for the same astronomical event: **Makara Saṁkrānti**. Today we learn what that is, astronomically.”

Core (20 min)

1. **Definition.** A *saṁkrānti* (transit) is a **moment**, not a day — the exact instant when the Sun’s ecliptic longitude crosses a multiple of 30°, leaving one *rāśi* and entering the next. The *day* on which this happens is named after the *rāśi* the Sun is entering.
2. **The 12 *saṁkrāntis* in order:**

#	<i>Saṁkrānti</i>	Sun enters	Approximate date
1	<i>Meṣa Saṁkrānti</i> (also <i>Mahā Viṣuva</i> / Tamil <i>Puthandu</i>)	Meṣa (Aries)	14 April
2	<i>Vṛṣabha Saṁkrānti</i>	Vṛṣabha (Taurus)	14 May
3	<i>Mithuna Saṁkrānti</i>	Mithuna (Gemini)	14/15 June
4	Karka Saṁkrānti	Karka (Cancer)	16/17 July
5	<i>Simha Saṁkrānti</i>	Simha (Leo)	16/17 August
6	<i>Kanyā Saṁkrānti</i>	Kanyā (Virgo)	16/17 September
7	<i>Tulā Saṁkrānti</i>	Tulā (Libra)	17/18 October
8	<i>Vṛścika Saṁkrānti</i>	Vṛścika (Scorpio)	16/17 November
9	<i>Dhanus Saṁkrānti</i>	Dhanus (Sagittarius)	15/16 December
10	Makara Saṁkrānti	Makara (Capricorn)	14/15 January
11	<i>Kumbha Saṁkrānti</i>	Kumbha (Aquarius)	12/13 February
12	<i>Mīna Saṁkrānti</i>	Mīna (Pisces)	14/15 March

3. **Two *saṁkrāntis* are especially significant:**

- **Makara Saṁkrānti** (~14 January) — entry into *Makara*. In the *nirayana* system, this is the traditional start of *uttarāyana*. Across India:
 - **Pongal** in Tamil Nadu (4-day harvest festival)
 - **Lohri** in Punjab and Haryana (bonfire eve)
 - **Bihu** in Assam (Magh Bihu)
 - **Maghi** in Punjab (Sikh new year)
 - **Uttarayan** in Gujarat (kite festival)
 - **Khichdi** in Uttar Pradesh and Bihar
 - **Suggi** in Karnataka
 - **Karka Saṁkrānti** (~16 July) — entry into *Karka*. In the *nirayana* system, this is the traditional start of *dakṣiṇāyana*. Celebrated as **Dakṣiṇāyana day** in parts of Maharashtra and as the **Rath Yatra** preparations in Odisha (loosely tied).
4. **Reference to the Bṛhat Saṁhitā** (Middle-band paraphrase):

Varāhamihira (6th c. CE), in his encyclopedic *Bṛhat Saṁhitā*, dedicates chapters to the *saṁkrāntis* — listing their dates, their seasonal markers, and (in his cultural context) their religious significance. The astronomical core — the Sun crossing a *rāśi*-boundary — is what we extract here. — paraphrased from *Bṛhat Saṁhitā*, chapter on solar transits (see [sources.md](#)).

Activity (15 min) — Activity 02 *Makara Saṁkrānti* explainer

See [activities/activity-02-makara-sankranti.md](#) for the full sheet. In summary:

- In small groups of 3–4, students prepare a 2-minute explanation of *Makara Saṁkrānti* answering:
 - What is the astronomical event?
 - What date does it fall on?
 - What regional festivals celebrate it?
 - What does it traditionally mark?
- Each group presents to one neighbouring group (pair-share). Last 3 minutes: one group presents to the full class.

Wrap-up (5 min)

- Exit prompt (verbal): “*Today is (today’s date). Which saṁkrānti is closest? Past or upcoming?*”
- Preview Day 7: “*Tomorrow — the BIG question. Why is Makara Saṁkrānti on 14 January and not on 21 December (the actual winter solstice)? It’s a 24-day gap. There’s a reason.*”

Student-facing content

Saṁkrānti = the Sun crosses into a new *rāśi*. There are 12 per year.

The two most important:

- **Makara Saṁkrānti** (~14 January) — Sun enters *Makara* (Capricorn). Marks the start of *uttarāyaṇa* in the *nirayana* (sidereal) system. Celebrated as Pongal / Lohri / Bihu / Maghi / Uttarayan across India.
- **Karka Saṁkrānti** (~16 July) — Sun enters *Karka* (Cancer). Marks the start of *dakṣiṇāyana*.

Each *saṁkrānti* is a calendar moment, named after the *rāśi* the Sun is entering. Local festivals attach to many of them.

Homework

- Add the 12 *saṁkrānti* dates to your *rāśi*-wheel — write the approximate date at each *rāśi*-boundary.
- If your family celebrates one of the *Makara Saṁkrānti* festivals (Pongal, Lohri, Bihu, etc.), ask one elder how they celebrate, and write 3 sentences. If your family doesn't celebrate any of them, write 3 sentences from a classmate's account.

Differentiation

- **One tier up:** Each *saṁkrānti* has a religious / cultural significance attached in traditional almanacs. List the 12 and their associated traditional names from a *pañcāṅga* or Wikipedia. Note: this is cultural-historical knowledge, not astronomy — keep the two separate.
- **One tier down:** Memorise just two — *Makara* (January) and *Karka* (July). That's enough for Day 7.

Teacher notes

- Resist the urge to “explain Day 7” early. Hold the puzzle: *Makara Saṁkrānti* on Jan 14 vs winter solstice on Dec 21 = 24-day gap. Let students sit with the gap; tomorrow we resolve it.
- Be ready for a religious-festival drift. Acknowledge briefly that these are real festivals; redirect to the astronomy.
- The Varāhamihira reference is paraphrased — don't pretend to quote.

Citation(s) used in this lesson

- **Varāhamihira, *Bṛhat Saṁhitā*** (6th c. CE) — paraphrased reference to the chapters on *saṁkrāntis*. See [sources.md](#).

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will distinguish the **sāyana** (tropical) and **nirayana** (sidereal) zodiacs, explain that the gap between them — the **ayanāṁśa** — is ~24° today, and link this gap to **precession** (the slow wobble of Earth's axis).

Materials

- Whiteboard
- Two transparent circles printed on overhead-projector film or paper-cutouts: each shows the 12 *rāśis* arranged around a circle
- A long printed strip showing the year (Jan to Dec) with the two solstice markers
- Student notebooks

Vocabulary introduced today

- *sāyana* (IAST: *sāyana*; lit. “with-ayana / tropical”) — the reference frame fixed to the equinoxes. The Sun enters *Meṣa* (Aries) on the **March equinox** by definition. The seasons stay locked to this system.
- *nirayana* (IAST: *nir-ayana*; lit. “without-ayana / sidereal”) — the reference frame fixed to the **fixed stars**. The Sun enters *Meṣa* when it physically passes a fixed point against the background stars.
- *ayanāṁśa* (IAST: *ayana-aṁśa*; lit. “precessional portion”) — the angular offset between the two systems, $\sim 24^\circ$ today.
- **Precession** — the slow wobble of Earth’s rotation axis. It traces a cone in space over $\sim 25,800$ years. Because of this wobble, the position of the equinoxes drifts among the stars at ~ 50 seconds of arc per year ($\sim 1^\circ$ per 72 years).

Lesson flow

Warm-up (5 min)

- Recall the puzzle: *Makara Saṁkrānti* is ~ 14 January but the winter solstice is ~ 21 December — a 24-day gap.
- Prompt: “24 days. Why 24?”
- Take 2 guesses. Park them. “Today we resolve it.”

Core (25 min)

1. Two ways to define the start of the zodiac.

Imagine the great circle of the ecliptic divided into 12 *rāśis*. **Where does *Meṣa* (the first *rāśi*) begin?**

- **Option A (*sāyana* / tropical):** *Meṣa* begins at **the March equinox point**. This makes the Sun’s position lock-step with the seasons. When the Sun is at 0° *Meṣa*, it’s the March equinox — always. Modern Western astronomy uses this.
- **Option B (*nirayana* / sidereal):** *Meṣa* begins at **a fixed point against the stars** (chosen near the star Spica / *Citrā* by Indian convention). When the Sun is at 0° *Meṣa*, it’s physically passing that star-anchored boundary. Indian traditional *jyotiṣa* uses this.

2. Why do they differ? Because of **precession**.

- Earth’s rotation axis isn’t perfectly fixed in space. It’s a slow wobble — like a spinning top tilting in a circle.

- One full wobble takes about **25,800 years**.
- The equinox point (where the Sun crosses the equator going north) is defined by the axis — so as the axis wobbles, the equinox point **drifts** against the background stars at about **50 seconds of arc per year**, or **1° per 72 years**.
- The drift goes *backwards* through the zodiac — from *Meṣa* into *Mīna*, then into *Kumbha*, and so on.

3. The size of the gap right now.

- The two systems were aligned approximately **1700 years ago** (in the early centuries CE).
- In that time, precession has drifted the equinox by **~24°** against the stars.
- So the *sāyana* zodiac and the *nirayana* zodiac are now **24° apart**.
- $24^\circ / 360^\circ \times 365.25 \text{ days} \approx 24.3 \text{ days}$. That's the gap.
- **24-day gap = 24° offset = ~1700 years of precession**. That's why *Makara Saṁkrānti* is 24 days *after* the winter solstice.

4. Demo with the two transparencies.

- Place the two *rāśi* circles on top of each other, both starting at the top.
- Slide one circle 24° clockwise. Now both circles still have *Meṣa*, *Vṛṣabha*, ... in order, but the boundaries are offset.
- “If the Sun is at 0° *Meṣa* in the bottom (*sāyana*) circle on March 21, where is it in the top (*nirayana*) circle?” — answer: it's at 0° *Mīna* + 6° (since it's offset 24° backward). It enters *Meṣa nirayana* ~24 days later, on ~April 14.

5. A summary table:

Event	<i>Sāyana</i> date	<i>Nirayana</i> date	Gap
Sun enters <i>Makara</i> / Capricorn	~21 December (= winter solstice)	~14 January (= Makara Saṁkrānti)	~24 days
Sun enters <i>Karka</i> / Cancer	~21 June (= summer solstice)	~16 July	~25 days
Sun enters <i>Meṣa</i> / Aries	~21 March (= March equinox)	~14 April	~24 days
Sun enters <i>Tulā</i> / Libra	~23 September (= September equinox)	~17 October	~24 days

6. Which is “right”? Neither — they're answering different questions.

- *Sāyana* answers: “Where is the Sun relative to the seasons (equinoxes / solstices)?” — used by modern astronomy and the civil calendar.
- *Nirayana* answers: “Where is the Sun against the actual background stars?” — used by traditional Indian almanacs.
- **Both are internally consistent**. They diverge because the equinox drifts; whichever frame you fix, the other one moves.

Activity (10 min) — Compare-contrast worksheet

(See [activities/activity-03-sayana-nirayana.md](#) for the full sheet.)

In pairs, students fill a two-column table:

Question	<i>Sāyana</i> answer	<i>Nirayana</i> answer
Sun in <i>Meṣa</i> on ...	~21 March	~14 April
Aligned with seasons?	Yes (by definition)	Drifts over centuries
Aligned with stars?	Drifts over centuries	Yes (by definition)
Used by modern astronomy?	Yes	No
Used by Indian <i>pañcāṅga</i> ?	No	Yes
What changes over millennia?	Star background drifts	Season alignment drifts

Wrap-up (5 min)

- Exit ticket: “Why is *Makara Saṁkrānti* on January 14 and not December 21? Answer in one sentence.”
- Acceptable answer: “Because the *nirayana* (sidereal) zodiac, used for *Makara Saṁkrānti*, is offset from the *sāyana* (tropical) zodiac, used for the solstice, by ~24° due to precession over ~1700 years.”
- Preview Day 8: “Tomorrow — build the project. Your *rāśi*-wheel will show BOTH systems.”

Student-facing content

Two zodiacs, one sky.

System	Anchored to	Used by
<i>Sāyana</i> (tropical)	The equinoxes	Modern astronomy, Western horoscopes
<i>Nirayana</i> (sidereal)	The fixed stars	Indian traditional almanacs

The two were aligned ~1700 years ago. Earth’s slow axial wobble — **precession** — has drifted them apart by ~24° since. That gap is called the **ayanāṁśa**.

$24^\circ \div 360^\circ \times 365.25 \text{ days} \approx \mathbf{24 \text{ days}}$. That’s why *Makara Saṁkrānti* (Jan 14, *nirayana*) is 24 days after the winter solstice (Dec 21, *sāyana*).

Homework

- Look up the *ayanāṁśa* value for today on any astronomy website or *pañcāṅga* app. Most sources will quote either the *Lahiri* value (the Government of India standard) or the *Krishnamurti* value. They differ by a few minutes of arc. Note both.
- Write 4 sentences: explain to a younger sibling why their newspaper horoscope sign (Western, *sāyana*) might be one *rāśi* “off” from a traditional *pañcāṅga* (Indian, *nirayana*).

Differentiation

- **One tier up:** When will the *sāyana* and *nirayana* zodiacs realign? Hint: when *ayanāṁśa* completes another $\sim 336^\circ$ ($= 25,800 \text{ yrs} \times 336/360$). Approximately when? (Answer: in $\sim 24,000$ years.) What was the *ayanāṁśa* in 1 CE? (Answer: very close to 0° .)
- **One tier down:** Just remember: ~ 24 days difference. *Sāyana* = with the seasons. *Nirayana* = with the stars. Don’t worry about precession mechanics yet.

Teacher notes

- This is the densest day of the module. Plan to take questions — and to defer some to next class. The full theory of precession is the Day 8 *senior* topic; here we just name it.
- Some students will be upset that “the calendar is wrong.” Reframe: neither calendar is wrong. Each is right *within its frame*. The drift is a real astronomical phenomenon.
- If you have only 35 minutes, drop the worksheet and use the time for questions and the demo.

Citation(s) used in this lesson

- Precession rate ($\sim 50''/\text{year}$, $\sim 25,800$ -year period) is a modern astronomy standard value (see *sources.md*). Lahiri *ayanāṁśa* is the Government of India standard adopted in the 1950s by the Calendar Reform Committee chaired by Meghnad Saha.

Day 8 — Project Session 1: Build the Rāśi-Wheel

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this session, every pair has a substantial draft of the *rāśi*-wheel project — labelled, coloured, with the four solstice/equinox markers and at least 6 of the 12 *saṁkrānti* dates filled in.

Materials

- The in-progress *rāśi*-wheel from Day 5 (one per pair)
- Coloured pencils, sketch pens, markers
- A4 / A3 chart paper for an optional larger wheel
- Glue, scissors, ruler
- The *sources.md* file printed for reference

- Project rubric (assessments/rubric.md) — printed and visible

Lesson flow

Warm-up (5 min)

- Recap the project brief (5 sentences): each pair builds a *rāśi*-wheel that shows BOTH the *sāyana* and *nirayana* dates for the four turning-points, the 12 *rāśi* names with English/Greek equivalents, and at least one paragraph linking it to axial tilt and seasons.
- Show the rubric (it should already be on the board). Quick Q&A: 2 minutes for any rubric clarifications.

Core (5 min)

The teacher explicitly walks through the “Meets” target for the project:

- 12 *rāśi* names in correct order ✓
- English / Greek equivalents matching ✓
- Approximate *nirayana* date ranges (e.g. Meṣa = 14 Apr – 14 May) ✓
- All four solstice / equinox markers labelled with both *sāyana* and *nirayana* dates ✓
- A 4-sentence written explanation of why we have seasons (axial tilt) ✓
- Citation to *Sūrya-siddhānta* OR *Bṛhat Saṃhitā* OR an NCERT chapter ✓

To **Exceed**, add ONE of: - A clearly-drawn precession diagram showing the wobble cone - A second wheel for Greek-origin star-name comparisons - A short paragraph on the cross-cultural Sanskrit-Greek-Babylonian zodiac correspondence

Build time (30 min)

Pairs work in silence first (10 min) to get the labels down, then collaborative (20 min) to add the explanatory text, colour, and diagrams.

Teacher circulates:

- For pairs stuck on labelling: hand them the printed reference table from Day 5.
- For pairs stuck on axial-tilt explanation: redirect to the Day 2 sketch in their notebook.
- For pairs racing ahead: ask “what would happen if the tilt were 0°?” or “how would a Babylonian astronomer label this wheel differently?”

Check-in (5 min)

Stop work at minute 40. Each pair holds up their current draft. The teacher does a quick visual scan and notes 2–3 pairs to circle back to first thing tomorrow.

- Exit prompt: “*What’s the ONE thing you still need to finish before tomorrow?*”
- Write it on a sticky note attached to the wheel.

Pair coaching prompts (for the teacher)

If you see ...	Try saying ...
One partner doing all the drawing	“Tell me what your partner is contributing. Where will their writing go?”
A wheel with English names only	“Where do the Sanskrit names go? Show me.”
Confusion about <i>sāyana</i> vs <i>nirayana</i>	“Pick the four turning-points. Write the December solstice date on one line. Now write the Makara Saṁkrānti date below it. They’re different — your job is to make that visible on the wheel.”
Beautiful art, no astronomy	“Where on the wheel does the reader learn that the seasons come from axial tilt? Add a sentence.”

Differentiation

- **One tier up:** Add a second concentric ring showing the Western horoscope (*sāyana*) date range for each *rāśi*. The reader should be able to compare the two rings at a glance.
- **One tier down:** Drop the *sāyana* / *nirayana* dual labelling. Just label one set of dates clearly. The *ayanāmśa* concept appears only in the written paragraph.

Teacher notes

- Some pairs will spend the entire 45 minutes colouring and lose the astronomy. Intervene at minute 20: “show me your written paragraph on axial tilt.” If it doesn’t exist yet, stop the colouring.
- A surprising number of pairs will get the *rāśi* order slightly wrong (especially *Tulā* ↔ *Vṛścika*). Catch this early.

Homework

- Finish at home what didn’t get done in class. Bring back tomorrow ready to refine, not start.
- If the pair worked separately, agree (over WhatsApp or in person) who finishes what.

Day 9 — Project Session 2: Cross-Cultural Comparison and Refinement

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this session, every pair has a presentation-ready *rāśi*-wheel including a cross-cultural comparison element (Greek / Babylonian / Gregorian) and a rehearsed 3-minute presentation script.

Materials

- The in-progress *rāśi*-wheel from Day 8
- Reference printout: Sanskrit / Greek / Babylonian zodiac names side by side (provided below in *Cross-cultural reference*)
- Stopwatch / phone timer
- The rubric ([assessments/rubric.md](#))

Cross-cultural reference

The 12 zodiac signs across three traditions:

#	Sanskrit (Indian, <i>rāśi</i>)	Greek / English	Babylonian (~700 BCE origins)
1	<i>Meṣa</i> (Ram)	Aries (Ram)	LU.ḪUN.GÁ (Hired Man)
2	<i>Vṛṣabha</i> (Bull)	Taurus (Bull)	GU.AN.NA (Bull of Heaven)
3	<i>Mithuna</i> (Twins)	Gemini (Twins)	MAŠ.TAB.BA (Great Twins)
4	<i>Karka</i> (Crab)	Cancer (Crab)	AL.LUL (Crab)
5	<i>Siṃha</i> (Lion)	Leo (Lion)	UR.GU.LA (Lion)
6	<i>Kanyā</i> (Maiden)	Virgo (Maiden)	AB.SIN (Furrow / barley ear)
7	<i>Tulā</i> (Scales)	Libra (Scales)	RIN (Balance)
8	<i>Vṛścika</i> (Scorpion)	Scorpio (Scorpion)	GIR.TAB (Scorpion)
9	<i>Dhanus</i> (Bow)	Sagittarius (Archer)	PA.BIL.SAG (Archer)
10	<i>Makara</i> (Sea-goat)	Capricorn (Sea-goat)	SUḪUR.MAŠ (Goat-fish)
11	<i>Kumbha</i> (Water-pot)	Aquarius (Water-bearer)	GU.LA (Great One / water-pourer)
12	<i>Mīna</i> (Fishes)	Pisces (Fishes)	DU.NU.NU (Tails)

The Indian, Greek, and Babylonian names are *strikingly similar* — the same animals (ram, bull, twins, crab, lion, scorpion, fishes) appear in all three. This is not accidental. The most likely historical explanation is **shared origin in Babylonian astronomy (~700 BCE)**, transmitted to Greece (post-Alexander, ~300 BCE) and to India (Hellenistic contact, last centuries BCE). The 12-fold ecliptic division and the imagery were absorbed into the Indian *jyotiṣa* tradition and merged with earlier Vedic star-lore (the 27 *nakṣatra* system, which is genuinely older and independently Indian).

The 27-*nakṣatra* system is distinctively Indian; the 12-*rāśi* system is the Babylonian-Greek-Indian common ground.

Lesson flow

Warm-up (5 min)

- Project the cross-cultural table on the board. Take 30 seconds to let students absorb.

- Prompt: “*What patterns do you see?*” — collect 2 observations.
- Likely observation: “they’re all about the same animals.” Confirm: yes — and this is a real historical clue about cultural exchange.

Core (5 min)

- One-paragraph summary on the board:
 - The 12-fold zodiac is a shared inheritance across Babylon, Greece, and India.
 - Indian *jyotiṣa* layered this on top of the older 27-*nakṣatra* system.
 - This means: when you study the *rāśis*, you are studying a 2700-year-old transcontinental mathematical-astronomical inheritance, not a purely Indian invention.
- This is **not** “Indians borrowed from the West” — Babylonian astronomy reached India before it reached Greece in some respects, and the Indian elaboration (the *nirayana* convention, the *ayanāṁśa* corrections) is a real intellectual contribution.

Refinement work (25 min)

Pairs refine their wheel:

- Add the cross-cultural element. Options:
 - A small inset showing the Babylonian name for ONE *rāśi* alongside the Sanskrit and Greek.
 - A single sentence on the wheel: “These 12 signs appear in similar form in Babylonian, Greek, and Indian astronomy — a 2700-year inheritance.”
 - A short comparative note on Cancer / Karka with a footnote on the Babylonian name.
- Finalise the written explanation paragraph (4–6 sentences) covering: axial tilt, the two *ayanas*, the four turning-points, the *sāyana* / *nirayana* distinction, and one citation.
- Decide who-says-what in the 3-minute presentation.

Rehearsal (8 min)

In groups of two pairs (4 students), pairs rehearse their 3-minute presentation in front of each other.

The listening pair times the speakers and gives one piece of feedback:

- ONE thing that came across clearly
- ONE thing the audience didn’t understand

Then they swap.

Wrap-up (2 min)

- Exit signal: each pair raises hand showing 1, 2, 3, 4, or 5 fingers — how ready they feel for tomorrow.
- The teacher notes any pair showing 1 or 2 and plans a 5-minute conference at the start of Day 10.

Differentiation

- **One tier up:** Each pair preparing a “stretch” closer: a 30-second factoid the audience will remember (e.g. “Precession means in ~13,000 years, the December solstice will be in *Karka*, not *Makara*”). Confirm the factoid is astronomically accurate before they include it.
- **One tier down:** Skip the Babylonian table; just mention “the zodiac is shared between India, Greece, and the older Babylonian tradition.” That’s enough for a “Meets” presentation.

Teacher notes

- A small but real risk: a pair invents a Sanskrit etymology to “explain” a Greek name. Catch this — don’t let invented etymology into a presentation.
- The Babylonian names contain Akkadian / Sumerian cuneiform transliterations (LU.ḪUN.GÁ, etc.). Don’t drill these — they’re for visual comparison, not memorisation.
- Lateness on the wheel is acceptable as long as the presentation is rehearsed. Better an unfinished but well-explained wheel than a beautiful unstructured one.

Homework

- Final touches. By tomorrow morning, the wheel should be:
 - ☐ Labelled in Sanskrit and English / Greek
 - ☐ Coloured legibly
 - ☐ Showing all four turning-points with dual *sāyana* / *nirayana* dates
 - ☐ Including at least one cross-cultural reference
 - ☐ Accompanied by a 4–6 sentence explanation
 - ☐ Ready for a 3-minute presentation by both partners

Day 10 — Final Assessment + Showcase

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min (extendable to 60 if pair count > 8)

Learning objective

By the end of this session, the module is closed: the summative quiz is sat, the *rāśi*-wheel projects are presented to peers, and reflection slips are submitted.

Materials

- Printed summative quiz, one per student (`quizzes/summative.md`)
- The rubric (`assessments/rubric.md`)
- A wall / pinboard for the *rāśi*-wheel display
- A stopwatch / timer
- Reflection slips (paper)

Lesson flow

Summative quiz (20 min)

- Students sit individually for the printed quiz.
- 15 minutes of work + 5 minutes of buffer.
- Closed book.
- Submit on the way out for the showcase.

Showcase (20 min)

- Pairs present their *rāśi*-wheel.
- 3 minutes per pair (strict). 1 minute for Q&A.
- Teacher marks on the rubric during each presentation (5 min slot per pair).
- Audience: the rest of the class. Each non-presenting student writes ONE positive comment on a slip for the presenting pair (collected at the end).

If you have more than 8 pairs and 20 minutes is tight, split into two simultaneous showcase clusters (4 pairs each, in different corners), with the teacher rotating between them.

Reflection (5 min)

Each student fills the reflection slip:

1. What is ONE thing you'll remember about the Sun's movement six months from now?
2. What is ONE thing that didn't click for you in this module — that you want to come back to?
3. (Optional) Was there a “wait — that's surprising” moment? What was it?

Submit before leaving.

What gets graded today

Item	Marks	Where it lives
Summative quiz	25	quizzes/ summative.md
Project artefact	30	rubric assessments/rubric .md
Presentation (Day 10)	10	rubric assessments/rubric .md
Total	65 (scaled to 40 for module total — see rubric)	

Reflection slips are not graded; they inform next module's design.

Teacher notes

- The 3-minute timer is strict. The discipline of compression is part of the learning.
- If a pair runs over, the teacher gently raises a hand at 3 minutes; if they keep going past 3:30, end the presentation politely.
- For pairs that struggled — don't grade harshly during the presentation. Mark on substance + clarity, not on stage presence.
- Pin all the wheels on the wall after the showcase. Leave them up for a week.

Wrap-up

- 30 seconds, teacher: *“Two weeks ago, none of you could name the 12 rāśis. Today, all of you can — and you understand why **Makara Saṁkrānti** is on January 14 and not December 21. That's astronomical literacy that took the Indian tradition centuries to build. You built it in 10 days. Well done.”*
- Closing prompt: *“Tomorrow morning, when you walk to school, notice which way the Sun is rising. You'll find yourself looking for it. That's the habit this module was after.”*

Citation(s) used in this lesson

- All citations were addressed in earlier days. No new citations introduced today.

Differentiation

- **One tier up:** Students who finish the quiz early can begin annotating their *rāśi*-wheel with one additional astronomical fact each — checked by the teacher before being added to the displayed wheel.
- **One tier down:** Students struggling with the quiz: allow a 5-minute extension if needed; do NOT delay the showcase.

Module 12 (Middle) — Formative Quizzes

Two parts: Part A on Day 3 (after *ayana* lesson), Part B on Day 5 (after *rāśis* lesson). Both are diagnostic — not graded. Hand back within 24 hours.

Part A — Day 3 (3 min, 5 questions)

Name: _____ Class: _____

1. Write the Sanskrit name for the **northward course** of the Sun.

2. Which date marks the start of *uttarāyaṇa*?

- a) 21 March
- b) 21 June

- c) 23 September
d) **21 December** ←

3. Fill in the blank: “The Sun’s noon altitude RISES during _____.” (Choose *uttarāyaṇa* or *dakṣiṇāyana*.)

4. True or false: India straddles the Tropic of Cancer. (T / F)

5. Why are seasons caused by axial tilt, NOT by Earth’s distance from the Sun? (One sentence.)

Part A — answer key

Q	Answer	Notes
1	<i>uttarāyaṇa</i>	Spelling tolerated if recognisable; accept “uttarayana” without diacritics.
2	d) 21 December	The winter solstice.
3	<i>uttarāyaṇa</i>	Northward course = Sun climbs higher each noon.
4	T	The Tropic of Cancer runs through Bhopal, Ranchi, Kolkata.
5	Tilt determines how directly sunlight hits us and how many hours of daylight we get. Earth’s distance from the Sun barely changes through the year (Earth is actually CLOSER to the Sun in January).	Accept any answer that names tilt + concentrating angle / day length, NOT distance.

Part B — Day 5 (5 min, 5 questions, mixed format)

Name: _____ Class: _____

1. (2 marks) Write the first FIVE *rāśis* in order:

_____ → _____ → _____ → _____ → _____

2. (3 marks) Match each *rāśi* to its English/Greek equivalent. Draw lines.

Meṣa	Capricorn
Karka	Aries
Tulā	Cancer
Makara	Scorpio
Vṛścika	Libra

3. (2 marks) About how many days does the Sun spend in each *rāṣi*?

- a) ~7 days
- b) ~15 days
- c) ~30 days ←
- d) ~60 days

4. (1 mark) Name the *rāṣi* that the Sun is in during the month containing your birthday (using the *nirayana* table). Use Day 5's table.

5. (2 marks, short answer) The Sanskrit *rāṣi* names look very similar to the Greek/English names (Bull, Twins, Crab, Lion, ...). Why?

Part B — answer key

Q	Answer	Mark scheme
1	<i>Meṣa</i> → <i>Vṛṣabha</i> → <i>Mithuna</i> → <i>Karka</i> → <i>Simha</i>	2 if all 5 in correct order; 1 if 3–4 correct; 0 otherwise.
2	<i>Meṣa</i> ↔ Aries, <i>Karka</i> ↔ Cancer, <i>Tulā</i> ↔ Libra, <i>Makara</i> ↔ Capricorn, <i>Vṛścika</i> ↔ Scorpio	3 if all 5 matched; 2 if 3–4 matched; 1 if 1–2 matched.
3	c) ~30 days	2
4	Whichever <i>rāṣi</i> matches the student's birth-month per the Day 5 table.	1 if correct; allow ± 1 <i>rāṣi</i> if the birthday is near a <i>saṃkrānti</i> boundary.
5	Because both the Indian and Greek zodiacs share a common origin in Babylonian astronomy (~700 BCE), with later cross-cultural contact during the Hellenistic period. The 12-fold ecliptic division and most of the animal symbolism is a shared inheritance.	2 if “shared origin / cultural exchange” is named with any further detail; 1 if just “they look similar because”.

Total Part B: 10 marks. Not added to summative.

What to do with the results

- A class average below 6/10 on Part B = significant gaps. Reteach the *rāśi* order during Day 6 warm-up (or assign quick recall homework).
- Individual students below 3/10 on Part B = pull aside before Day 7's *sāyana/nirayana* lesson for a 5-minute one-on-one (otherwise Day 7 will not land).

Module 12 (Middle) — Summative Quiz

Time: 20 min · **Marks:** 25 · **Closed book**

Instructions: Write your name and class. Answer all questions. Section D includes one mark for clarity of writing. No calculators needed.

Name: _____ **Class:** _____

Section A — Multiple choice ($5 \times 1 = 5$ marks)

1. Earth's axial tilt is approximately:

- a) 0°
- b) 11.5°
- c) **23.5°** ←
- d) 45°

2. *Uttarāyaṇa* refers to:

- a) The Sun's daily motion east to west
- b) **The Sun's half-year course in which its noon altitude rises (~Dec 21 → ~Jun 21)** ←
- c) The new moon
- d) The summer solstice

3. Which two *saṁkrāntis* mark the start of the two *ayanas*?

- a) Meṣa Saṁkrānti and Tulā Saṁkrānti
- b) Mithuna Saṁkrānti and Dhanus Saṁkrānti
- c) **Makara Saṁkrānti and Karka Saṁkrānti** ←
- d) Kanyā Saṁkrānti and Mīna Saṁkrānti

4. The current *ayanāmśa* (offset between *sāyana* and *nirayana* zodiacs) is approximately:

- a) 0°
- b) 12°
- c) **24°** ←

d) 90°

5. True / false (circle one): The *sāyana* zodiac is “right” and the *nirayana* zodiac is “wrong.” T / F ←

Section B — Short answer ($4 \times 2 = 8$ marks)

6. Write the names of the four turning-points of the year (two solstices and two equinoxes), with approximate dates.

7. List the 12 *rāśis* in order.

8. Why does the Northern Hemisphere have summer in June and not in December, even though Earth is FARTHER from the Sun in July? Answer in 2 sentences.

9. Define *saṁkrānti*.

Section C — Diagram ($1 \times 5 = 5$ marks)

10. Draw a labelled diagram showing Earth’s orbit around the Sun. Mark the four positions corresponding to:

- June solstice
- September equinox
- December solstice
- March equinox

For each, show: - (a) Earth’s axis tilt (same direction at all four positions) - (b) Which date - (c) Whether the Northern Hemisphere is in summer / winter / equinox

[draw in the space below – full page]

Section D — Open response (1 × 7 = 7 marks; 1 of which is for clarity of writing)

11. Explain — to a parent who doesn't know astronomy — why **Makara Saṁkrānti** is celebrated around **14 January**, when the **winter solstice** is on **21 December**. Use 5–7 sentences. Include the terms *sāyana*, *nirayana*, *ayanāṁśa*, and *precession*.

Answer key (teacher copy)

Q	Answer	Mark scheme
1	c) 23.5°	1
2	b) The Sun's half-year course in which its noon altitude rises	1
3	c) Makara Saṁkrānti and Karka Saṁkrānti	1

Q	Answer	Mark scheme
4	c) 24°	1
5	F	1; both are right within their own frames
6	March equinox ~21 Mar; June solstice ~21 Jun; September equinox ~23 Sep; December solstice ~21 Dec	2 if all 4 with approximate dates; 1 if 2–3; 0 if 0–1.
7	Meṣa, Vṛṣabha, Mithuna, Karka, Siṁha, Kanyā, Tulā, Vṛścika, Dhanus, Makara, Kumbha, Mīna	2 if all 12 in correct order; 1 if 8–11; 0 otherwise. Accept missing diacritics if names are recognisable.
8	Because in June the Northern Hemisphere is tilted TOWARD the Sun — sunlight hits at a steeper angle and days are longer. Distance from the Sun affects temperature by less than 1%; axial tilt is the dominant factor for seasons.	2 if tilt + (angle OR day length) named, AND distance is correctly minimised. 1 if only tilt named.
9	The moment the Sun crosses from one <i>rāśi</i> into the next. There are 12 <i>saṁkrāntis</i> per year.	2 if both definition and “12 per year” present. 1 if only definition.
10	Earth at 4 positions, axis tilted in the SAME direction at each position (key check), correct seasonal assignments, dates labelled.	5: 1 for orbit shape; 1 for axis tilt drawn consistently; 1 for solstice dates correct; 1 for equinox dates correct; 1 for seasonal assignment correct.
11	Explanation should include: (a) <i>sāyana</i> (tropical) zodiac is anchored to the equinoxes; the winter solstice marks Sun entering <i>Makara</i> in <i>sāyana</i> ; (b) <i>nirayana</i> (sidereal) zodiac is anchored to fixed stars; Makara Saṁkrānti marks Sun entering <i>Makara</i> in <i>nirayana</i> ; (c) the two were aligned ~1700 years ago; (d) precession (Earth’s axial wobble, ~50″/year) has drifted them apart by ~24°; (e) 24° in solar motion = ~24	7: 1 for <i>sāyana</i> defined; 1 for <i>nirayana</i> defined; 1 for precession named/described; 1 for <i>ayanāṁśa</i> used in context; 1 for 24° → 24 days conversion mentioned; 1 for “neither is wrong” framing; 1 for clarity of writing.

Q	Answer	Mark scheme
	days, explaining the date gap.	

Total: 25 marks.

Grading bands (for project combined, total 65 scaled to 40):

- 32–40: Exceeds
- 24–31: Meets
- 16–23: Approaches
- 0–15: Beginning

Homework

Homework — Movement of the Sun (Middle)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sun’s Yearly Path** · Write 2 sentences about today’s concept. (~10 min)
2. **Day 2 — Axial Tilt and the Seasons** · Write 2 sentences about today’s concept. (~10 min)
3. **Day 3 — Uttarāyaṇa and Dakṣiṇāyana** · Write 2 sentences about today’s concept. (~10 min)
4. **Day 4 — Equinoxes — Viṣuva** · Write 2 sentences about today’s concept. (~10 min)
5. **Day 5 — Twelve Rāsis** · Write 2 sentences about today’s concept. (~10 min)
6. **Day 6 — Saṁkrānti Moments** · Write 2 sentences about today’s concept. (~10 min)
7. **Day 7 — Sāyana vs Nirayana** · Write 2 sentences about today’s concept. (~10 min)
8. **Day 8 — Precession of the Equinoxes** · Write 2 sentences about today’s concept. (~10 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Write 2 sentences about today’s concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today’s concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — Movement of the Sun (Middle)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 200-word response.

1. In 3–4 sentences, explain Movement of the Sun to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Movement of the Sun with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Movement of the Sun? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 01 — Shadow-Stick (Gnomon) Observation

Module: Movement of the Sun · **Band:** Middle · **Day:** 1 · **Time:** 15 min (within Day 1's 45 min)

Type

Outdoor / classroom-window observational science

Goal

Each pair measures and records the position of a gnomon's shadow at two different times, demonstrating the Sun's apparent daily motion.

Materials per pair

- One 30 cm wooden dowel, pencil, or chopstick (the **gnomon**)
- One A4 piece of stiff cardboard or chart paper (the **dial**)
- A small lump of clay or putty (to fix the gnomon upright)
- A compass (one per pair, or shared between two pairs)

- A ruler
- A pencil and a pen
- A stopwatch / phone timer for the second reading

Set-up (teacher, before class)

- Identify a sunny outdoor spot or an indoor location where direct sunlight enters through a window for at least 20 minutes during the lesson.
- Cut cardboard sheets in advance.
- Pre-mark a small N arrow on each cardboard (or instruct students to do this with the compass).

Instructions (student-facing — printable handout)

Shadow-stick observation — work in pairs.

Step 1 (3 min). Fix the gnomon upright on the cardboard with the clay. Use the compass to mark NORTH on the cardboard. Label the four cardinal directions (N, E, S, W).

Step 2 (1 min). Note the time on your stopwatch. Write it on the cardboard.

Step 3 (1 min). Trace the gnomon's shadow with a pencil. At the tip of the shadow, mark a small dot and write "T1" next to it.

Step 4 (10 min). WAIT — do not touch the gnomon or cardboard. While waiting, work in your notebook on the following questions: - Which direction is the shadow pointing? (Mostly opposite the Sun's position — so if Sun is in the east, shadow points west.) - Predict: in 10 minutes, will the shadow have moved CLOCKWISE or ANTICLOCKWISE around the gnomon? (Hint: as the Sun moves east-to-west, the shadow moves west-to-east — but it sweeps around the gnomon. From above the gnomon, the shadow rotates *clockwise* if you're in the Northern Hemisphere.)

Step 5 (1 min). At the 10-minute mark, trace the new shadow. Mark its tip with a "T2" dot.

Step 6 (5 min). Measure: how many degrees did the shadow rotate? (Use a protractor; the angle from T1 to T2 around the gnomon base.) - Expected: about 2.5° in 10 minutes (since the Sun moves 360° in ~ 24 hours = $15^\circ/\text{hour}$ = $2.5^\circ/10$ min).

Worksheet (one per pair)

Reading	Time	Shadow length	Shadow direction (which way it points from the gnomon)
T1		cm	
T2		cm	

Angle between T1 and T2 (around the gnomon base): _____ degrees

Predicted angle: 2.5° per 10 minutes. **Did your observation match?** ☐ Yes ☐ No (off by _____ $^\circ$)

Reflection (one sentence): Which direction did the Sun move while you were watching?
(Answer: from your earlier position to a later position — from east toward south toward west.)

Discussion prompts (last 3 min, whole class)

- Whose shadow rotated faster or slower than 2.5° ? Why? (Errors in timing, sloppy tracing, gnomon not vertical.)
- If we did this on the December solstice instead of today, would the shadow be longer or shorter? (Answer: depends on time of year; longer in winter, shorter in summer at noon.)
- If we drew shadows at 9 AM, noon, and 3 PM on the same day, what shape would the three tips trace? (Answer: a curve — not a straight line. At the equinox, it's a straight line; at solstices, a curve that bends away from north.)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners holding the cardboard steady, one tracing, one timing; quiet observation	"The shadow has moved!" / "It looks shorter than 10 minutes ago"
Needs intervention	Gnomon falling over; cardboard moved during the 10-min wait; one student looking at the Sun	Silence; or "I can't see anything" because the cardboard was moved

Safety / sensitivity

- **CRITICAL: Never look at the Sun directly.** Watch the shadow. Anyone caught glancing at the Sun is sent indoors for the rest of the activity. This is non-negotiable.
- If the schoolyard is unsafe for outdoor work, do this at a sunny window.
- Wear hats if outdoors for more than 15 minutes. Sunscreen if available.

Differentiation

- **Tier up:** Predict the shadow length you'd expect from your gnomon ($\text{length} \times \tan$ of the Sun's zenith angle). The Sun's zenith angle today (at noon) $\approx |\text{latitude} - \text{solar declination}|$. Compute and compare with measurement.
- **Tier down:** Just observe — no measurement. Note: "shadow moved from here to here."

Clean-up

- Take the cardboard back to the classroom — students keep their dials in their workbooks.
- Return the gnomons to the teacher.
- Compasses returned.

Activity 1 — Sunrise Direction Tracking

Module: Movement of the Sun · **Band:** Middle · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Movement of the Sun through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

compass app, classroom window, notebook

What students do

Over 5 days, students note the compass bearing of sunrise each morning (from the same window). They graph the bearing and predict how it will change in 3 months.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Makara Saṃkrānti Explainer

Module: Movement of the Sun · **Band:** Middle · **Day:** 6 · **Time:** 15 min (within Day 6's 45 min)

Type

Small-group prepared explanation; cross-cultural discussion

Goal

Each small group prepares and delivers a 2-minute explanation of *Makara Saṁkrānti* that distinguishes the **astronomy** (Sun's transit into *Makara rāśi*) from the **regional festivals** that attach to it.

Materials per group

- One A4 sheet — the “Saṁkrānti explainer card”
- Coloured pencils / sketch pens
- Map of India (printed, one per group, with state outlines visible)
- Reference printout: list of regional festival names tied to Makara Saṁkrānti

Set-up

- Form groups of 3–4. Mix students from different family backgrounds where possible.

Instructions (student-facing — printable handout)

Makara Saṁkrānti explainer — work in your group.

Prepare a 2-minute explanation. Your group will present it to one neighbouring group.

Your explanation must answer **all five questions** below:

1. **What astronomical event is *Makara Saṁkrānti*?** (*Answer: the saṁkrānti* — moment of transit — when the Sun enters the rāśi of Makara (Capricorn) in the nirayana / sidereal system. It happens around 14 January.*)*
2. **What does it traditionally mark?** (*Answer: the start of uttarāyaṇa* — the Sun's northward course — in the nirayana reckoning. It's traditionally considered auspicious in many Indian cultures.*)*
3. **What regional festivals share this date?** (*At least four of: Pongal, Lohri, Bihu, Maghi, Uttarayan, Khichdi, Suggi, Magha Sankranti. Mark them on the map of India.*)
4. **What is one common practice across these festivals?** (*Likely answers: harvest celebration, kite-flying, sesame-jaggery sweets, bonfires. Take whichever you actually know; don't invent.*)
5. **What is the astronomical difference between Makara Saṁkrānti and the winter solstice?** (*Answer: the winter solstice is December 21 in the sāyana* system. Makara Saṁkrānti is January 14 in the nirayana system. The ~24-day gap is the ayanāṁśa caused by precession over ~1700 years. The two were aligned ~1700 years ago.*)*

Worksheet (the explainer card)

Question	Your group's answer (in 1–2 sentences)
1. What astronomy?	
2. What does it mark?	

Question	Your group's answer (in 1–2 sentences)
3. Regional festivals	
4. Common practice	
5. Sāyana vs Nirayana gap	

Presentation order (within the group)

Decide:

- Who speaks Q1 and Q2 (the astronomy half)
- Who speaks Q3 and Q4 (the cultural half)
- Who speaks Q5 (the comparison)
- Who hands the card to the listening group at the end

Pair-share format (last 5 min)

- Two groups face each other.
- Group A presents (2 min).
- Group B asks one question and offers one positive comment.
- Swap roles.

Discussion prompts (whole class, last 3 min)

- Did anyone present a festival you'd never heard of?
- Did any group accidentally describe the festival without describing the astronomy? (This is a common mistake. Acknowledge gently.)
- The 24-day gap is real and important. Without it, *Makara Saṁkrānti* would BE the winter solstice. Why isn't it?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Group huddled, distributing speaking parts, marking the map	"I'll take Q3 since I know Pongal" / "Let me read out Q5"
Needs intervention	One student dominating; group treating it as a festival presentation only	"It's just a festival, that's all" — redirect: "Yes, but WHAT astronomical event triggers the festival?"

Safety / sensitivity

- Some students may feel personally attached to one of these festivals; others may not celebrate any. Make clear: we are not ranking festivals or implying that one region's celebration is more "Indian" than another's. Each is rooted in the same astronomical event.

Differentiation

- **Tier up:** Group adds a *Karka Samkrānti* equivalent — what astronomically? What festivals? (Hint: there's no large pan-Indian festival on July 16 the way there is on January 14. Why might that be?)
- **Tier down:** Skip Q5. Answer Q1–Q4 only. The *ayanāṁśa* question can be revisited individually.

Clean-up

- Hand the explainer card to the teacher.
- Maps go back in the materials box.

Activity 03 — Sāyana vs Nirayana Compare-Contrast

Module: Movement of the Sun · **Band:** Middle · **Day:** 7 · **Time:** 10 min (within Day 7's 45 min)

Type

Paired written compare-contrast

Goal

Each pair fills a two-column table comparing the *sāyana* (tropical) and *nirayana* (sidereal) zodiacs across six dimensions, demonstrating clear understanding of why they differ.

Materials per pair

- One printed worksheet (next page)
- A pen
- The Day 7 board diagram (or notebook copy of it)

Set-up

- Distribute one worksheet per pair. Pairs can be the same as Days 5–6 working pairs.

Worksheet — Sāyana vs Nirayana

Pair name: _____ & _____ **Date:** _____

Fill in the table. Each cell should be one sentence.

Question	<i>Sāyana</i> (tropical)	<i>Nirayana</i> (sidereal)
1. The Sun enters <i>Meṣa</i> (Aries) on what date?		
2. The Sun enters <i>Makara</i> (Capricorn) on what date?		

Question	<i>Sāyana</i> (tropical)	<i>Nirayana</i> (sidereal)
3. Fixed to what reference?		
4. What changes over centuries?		
5. Used by modern astronomy (Y/N)?		
6. Used by traditional Indian <i>pañcāṅga</i> (Y/N)?		

Short answer (3–4 sentences below the table):

In your own words, explain why both systems are “right” for their own purposes. Use the term *ayanāmśa* somewhere in your answer.

Answer key (teacher copy)

Question	<i>Sāyana</i>	<i>Nirayana</i>
1. Sun enters Meṣa	~21 March (the spring equinox by definition)	~14 April (when the Sun crosses the sidereal <i>Meṣa</i> boundary)
2. Sun enters Makara	~21 December (the winter solstice by definition)	~14 January (Makara Saṁkrānti)
3. Fixed to what?	The equinoxes (the points where Sun crosses celestial equator)	A specific point against the fixed stars (chosen near <i>Citrā</i> / Spica)
4. What changes?	The position of the equinoxes against the stars (precession)	The alignment with the seasons (the same precession, seen the other way around)
5. Modern astronomy uses?	Yes — civil calendars, weather science, Western horoscopes	No (modern astronomy uses <i>sāyana</i> for civil dates; uses pure star coordinates for catalogues, but not the <i>rāśi</i> boundaries)
6. Indian <i>pañcāṅga</i> uses?	No (almost universally)	Yes

Short-answer rubric: the answer should include:

- The point that *neither* is right — they’re answering different questions.
- The point that the *seasons* and the *stars* drift relative to each other because of precession.

- The word *ayanāṁśa* used in context (the offset, $\sim 24^\circ$ today).
- A consequence: e.g. “Western horoscopes use *sāyana*, Indian *pañcāṅgas* use *nirayana*, so a person’s ‘Sun sign’ may be one *rāśi* different in the two systems.”

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair pointing at the Day 7 diagram, debating one cell at a time	“Wait — is <i>Makara</i> on Dec 21 or Jan 14?” / “Both, in different systems!”
Needs intervention	Both cells filled with the same answer; or one student copying from the board without discussion	Silence; one-word answers

Discussion prompts (whole-class, last 3 min if time)

- Did any pair disagree on Q5 (modern astronomy) or Q6 (*pañcāṅga*)? Why?
- A challenge: if Earth’s axis suddenly stopped precessing tomorrow, how long would it take for *Makara Saṁkrānti* to drift to coincide with the winter solstice? (*Trick question: it wouldn’t drift back — it would just stay 24° offset forever. The drift CAUSED the gap, and stopping precession would just freeze the gap.*)

Differentiation

- **Tier up:** Compute the year in which the two systems WILL re-align by precession. Hint: $\sim 25,800$ years $- 1700$ years = $\sim 24,100$ years from now.
- **Tier down:** Fill only Q1, Q2, Q5, Q6. The “what changes” and the short-answer can be skipped.

Project Brief

Project Brief — Personal Solar Calendar

Module: Movement of the Sun · **Band:** Middle

What you will do

Make a one-page solar calendar for the current year: mark the two solstices, two equinoxes, all 12 *saṁkrānti* dates, plus your own birthday with its *rāśi*.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (300 words (Middle)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 2 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Project Rubric — Module 12 Movement of the Sun (Middle)

Project marks: 30 (out of 40 total when combined with presentation 10).

Criteria

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Conceptual accuracy	All 12 <i>rāśis</i> in correct order with both Sanskrit and English/Greek names. Both <i>sāyana</i> and <i>nirayana</i> dates shown for all 4 turning-points. Explanation paragraph correctly links axial tilt to seasons.	All 12 <i>rāśis</i> in order; English/Greek correct; 4 turning-points labelled with at least one date system. Minor imprecision OK.	10–11 <i>rāśis</i> correct OR turning-points missing or mislabelled.	Fewer than 10 <i>rāśis</i> , OR major errors (e.g. wrong solstice direction).	× 2
Use of sources	At least one citation (Sūrya-siddhānta, Bṛhat Saṃhitā, or an NCERT chapter) named in the project AND used correctly in context. Bonus: a paraphrased point from the cited source.	One citation named correctly somewhere on the project.	A source mentioned vaguely (“an old Indian text says...”) without specific reference.	No source cited.	× 2
Sāyana / Nirayana distinction	Both systems labelled on the wheel; the <i>ayanāṁśa</i> offset shown numerically; one sentence explaining why the two differ.	Both systems mentioned; the ~24-day gap shown for at least one turning-point (e.g. Dec	One system shown; the other only mentioned in text.	Only one system shown; the other not mentioned.	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
Cross-cultural element	Compares Sanskrit names to at least TWO other traditions (Greek + Babylonian, or Greek + Gregorian calendar). Shows awareness that the 12-zodiac is a shared inheritance, not exclusively Indian.	21 vs Jan 14). Compares Sanskrit names to one other tradition (typically Greek).	Mentions the comparison only in passing.	No cross-cultural element.	× 1
Originality of expression	The wheel shows the pair's own design choices — a fresh layout, a surprising visual element, a turn of phrase in the explanation.	Clearly the pair's own work; not copied from a single online template.	Clearly based on a single online template; little independent thinking.	Clearly copied.	× 1
Communication / craft	Visuals, labelling, and explanation are clean, deliberate, and serve the message. Sanskrit IAST is correctly typed. Spelling	Communicates clearly. One or two craft issues.	Message comes through despite some craft issues.	Hard to follow; multiple craft issues impede understanding.	× 1

Criterion	4 — Exceeds	3 — Meets	2 — Approaches	1 — Beginning	Weight
	correct.				

Score calculation

Project score = (sum of criterion scores × weights) × (30 / 32) = sum × 0.9375

Maximum raw weighted score: 32 (eight weighted points × max 4). Scale to 30.

Presentation marks (10, on Day 10)

Criterion	4	3	2	1	Weight
Within time (3 min)	Within 2:45–3:00	Within 2:30–3:15	Within 2:15–3:30	Outside that range	× 1
Both partners contribute (or solo presents whole)	Equal contribution; smoothly handoffs	Both partners speak but unequally	One partner dominates	One partner silent	× 1
Clarity and audibility	Confident, audible to the back of the room, clear pace	Audible, mostly clear	Audible only in front rows; some mumbling	Mostly inaudible	× 0.5

Maximum: 10.

Self-reflection mark (folded into project score)

The 3 reflection questions on Day 10 are not separately scored — but if the reflection is missing or perfunctory, the **Originality** mark drops by 1.

Common decision points

- **Group of 3 vs pair:** Acceptable, but the third member must have a clearly distinct sub-role (e.g. visuals + script + cross-cultural research). Marks are equal across the group unless one member visibly didn't contribute.
- **Late submission:** -2 marks per school day, capped at -6.

- **Originality dispute:** If a wheel is clearly copied from an online template (some templates are easily recognised), the project goes to 0 on **Use of sources** AND on **Originality**. A 1-day extension is offered for a redo.

Marking process

- Mark in real-time during presentation (5 minutes per pair: 3 present + 1 Q&A + 1 marking).
- Write 1-sentence written feedback after each presentation finishes.
- Return marked rubrics within 2 school days.

Examples — “Meets” vs “Exceeds”

- **Meets:** All 12 *rāśis* labelled; the Sanskrit-Greek correspondence shown; the four turning-points marked with *nirayana* dates; explanation paragraph mentions axial tilt; *Sūrya-siddhānta* cited; paper is neatly drawn.
- **Exceeds:** Same as above PLUS both *sāyana* and *nirayana* dates shown for all four turning-points; the *ayanāṁśa* (~24°) is shown visually as a small offset between two concentric rings; the explanation paragraph correctly mentions precession; the Babylonian column appears in a small inset comparing names.